

Local Control over Market Integration

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Abstract

Market integration directs resources toward higher-value uses, but the resulting spatial reallocation of economic activity can generate local spillovers. Communities may anticipate these spillovers and exert protectionist local control, limiting gains from trade. We exploit differential exposure to the integration of the world's largest water market to document local impacts that generate incentives for protectionism and quantify how much communities value retaining irrigation. We find that a 10% decrease in local irrigation reduces household incomes by 3%, agricultural employment by 7%, and population by 5%. We use a dynamic residential choice model to recover a distribution of household willingness to pay to retain the broader local benefits of irrigation. Aggregating household valuations by location, median willingness to pay for irrigation is AUD \$119/ML, with 13% of locations valuing water above the average market price of \$235/ML. We combine these valuations with estimates of agricultural productivity to quantify how community veto rights would limit trade. Under asymmetric information, where transacting farmers must compensate communities for their privately known value of water, only 49% of trades that increase farmer surplus remain feasible. Compared with community veto rights, uniform compensation requires larger payments to preserve comparable levels of trade while failing to fully compensate many communities for their losses. Information, targeting, and the surplus available for compensation all shape the extent to which local control limits trade.

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1 Introduction

As climate change intensifies water scarcity, water markets offer a powerful tool for adaptation by reallocating scarce supplies toward higher-value uses. Yet, globally less than 1% of surface water entitlements are traded (Rafey, 2023, Bruno & Jessoe, 2024). Jurisdictions often restrict water transfers to protect against the local economic consequences of water leaving the region, substantially limiting trade (Hanak, 2005, Grafton et al., 2011). This local control of trade flows ranges from explicit protections for communities losing water,¹ to local water authorities restricting sales outside their boundaries.² The demand for these controls reflects a central conflict for water markets: the same reallocation that generates gains from trade can move economic activity away from communities that historically depended on irrigation, creating local incentives to block trade.

Local resistance is a recurring challenge to market integration. Import competition can harm communities exposed to foreign competition, development can impose costs on nearby residents, and resource markets can move economic activity away from places that historically depended on it (Autor et al., 2013, Bartik et al., 2019, Currie et al., 2015). These local spillovers can create demand for protection from market forces, ranging from trade barriers to restrictions on development and resource transfers. A central question for market design is therefore how markets perform when affected communities have the power to block otherwise mutually beneficial trades.

In this paper, we study a sweeping 2007 reform that integrated Australia’s Murray–Darling Basin into the world’s largest water market, generating large and persistent spatial reallocation of irrigation water across communities. This setting provides a unique opportunity to estimate the key economic forces underlying the conflict between market integration and local control. Water is a measurable productive input whose movement across locations can be observed, allowing us to measure both its value to affected communities and the private agricultural profits that motivate trade. Because historical water allocations and limited trade left irrigation imperfectly aligned with agricultural productivity, the reform differentially exposed communities to water reallocation according to their agricultural suitability. We measure this suitability using predetermined soil characteristics, whose interaction with the timing of reform provides a difference-in-differences instrument for changes in local irrigation water use. We exploit this variation to quantify both the specific local consequences of water reallocation and households’ overall valuation of retaining irrigation.

We first estimate how reform-induced water reallocation affects specific local outcomes that may generate incentives for protectionism. We find that these effects extend well beyond the farmers directly participating in water markets. A 10% reduction in local irrigation water use reduces population by 4.5%, household income by 3.1%, housing costs by 5.1%, and the stock of long-term

¹California law, for example, conditions access to public conveyance capacity on transfers not “unreasonably affecting the overall economy” of the source county (California Water Code §1810). Some counties have adopted additional restrictions on water exports to protect agricultural economies (Hanak, 2005).

²Local institutional restrictions are widespread: before Australia’s water-market reforms, irrigation districts could restrict water trades (Australian Competition and Consumer Commission, 2009); similar restrictions remain common in the American West (Culp et al., 2014); and Spain subjects transfers to basin-level institutional approval (Calatrava & Martínez-Granados, 2015).

residents by 3.4%. Predicted post-reform changes in irrigation are unrelated to local outcomes prior to the reform, supporting the identifying variation used throughout the paper (Alsan et al., 2025). Consistent with these local consequences generating opposition to reallocation, voters shift against the party responsible for the reform, and we also find evidence of increased deaths of despair. These results establish economically meaningful effects along several important margins, but the consequences of water reallocation are difficult to enumerate and value individually. For local control, the key economic object is instead overall value a community places on retaining irrigation. Its magnitude and heterogeneity, and its relationship with the productive gains from reallocation, determine which trades local control would block.

We recover this broader willingness to pay to retain the local benefits of irrigation with a dynamic residential choice model (Almagro & Dominguez-Iino, 2025, Davis et al., 2023). Using households' observed location decisions from 2006 to 2021, the model allows households to value income, housing costs, their own attachment to place, and the rootedness of the surrounding community. Water can affect household welfare through each of these margins as well as through unobserved local amenities, yielding an encompassing valuation without requiring us to enumerate every consequence of water reallocation. Reform-induced changes in irrigation identify the average effect of water on household utility. Repeated observations of each community after the reform then reveal persistent differences in this effect across locations, identifying the distribution of community values even when location-specific causal effects cannot be estimated individually.

We find that these local valuations are economically substantial. Aggregating household valuations by location, median willingness to pay for irrigation is AUD \$119/ML, with 13.5% of locations valuing water above the average market price of \$235/ML. We estimate that local irrigation's effect on household income accounts for 24.2% of median value. Local valuations are also highly heterogeneous, ranging from approximately \$71/ML at the 25th percentile to \$185/ML at the 75th percentile. The magnitude and heterogeneity of these valuations determine the compensation communities may require to permit water transfers. Whether that compensation can be supported by gains from trade also depends on the spatial correlation between community valuations and the private value of water that drives market reallocation.

To measure the private agricultural profits that motivate trade, we adapt the agricultural production model and estimates of Rafey (2023) to our geographic units of analysis. This yields location-specific marginal productive values that we compare with the value of irrigation to local residents. Productive values are also highly heterogeneous, with a median of approximately \$265/ML and 10th and 90th percentiles of \$127/ML and \$840/ML. Despite substantial heterogeneity in both values, their spatial correlation is close to zero and only weakly positive. Consistent with the allocative objective of the reform, water moves toward locations with higher productive values, while community valuations do not predict where water moves.

Our estimates reveal three features of the joint distribution that shape the feasibility and design of compensation. First, community values are substantial relative to productive values, so compensation can absorb a meaningful share of the gains from trade. Second, community values are highly

heterogeneous, increasing the value of targeting transfers across communities. Third, community and productive values exhibit a correlation of just 0.04: the market reveals where surplus can be created, but provides little information about where that surplus is needed for compensation.

We use these estimates to study how water markets perform when communities have the right to block transfers that reduce local irrigation. A trade can proceed only if the gains it creates are sufficient to secure the selling community’s consent. We consider two features that determine how restrictive these veto rights are. First, we vary the surplus available to secure consent, comparing trades financed only by the gains to transacting farmers with those that can also draw on gains accruing to the buying community. Second, we vary information about the selling community’s value, comparing a full-information benchmark with an asymmetric-information environment in which this value is privately known.

Under full information, gains to transacting farmers allow 68.1% of otherwise profitable trades to proceed while fully compensating selling communities. Allowing gains to the buying community to contribute increases this share to 89.7%. Private information makes local control substantially more restrictive. When transacting farmers make optimal take-it-or-leave-it offers to selling communities with privately known values, only 49.1% of trades proceed using private gains alone; allowing the buying community to contribute increases this share to 66.2%. Finally, we compare community veto rights with a uniform compensation scheme, providing an empirical analogue to the classic distinction between Coasian bargaining and Pigouvian instruments (Coase, 1960, Pigou, 1920). Uniform compensation requires substantially larger payments to preserve comparable levels of trade while leaving more affected communities undercompensated. These results show that the market consequences of local control depend on the information available to target compensation and the set of gains that can be mobilized to secure community consent.

Our results make three contributions, concerning the political economy of market integration, the design of water markets, and the measurement of heterogeneous local welfare effects.

A large literature shows that market integration and reallocation toward higher-value uses can impose large and persistent losses on particular workers and places, generating political demand for protection (Autor et al., 2013, Fajgelbaum et al., 2020, Kuziemko et al., 2026). This literature has largely studied integration in product and labor markets rather than markets that reallocate scarce resources across places. Resource markets create a distinct problem: trades generate private gains by moving productive inputs between communities, while the resulting local effects fall partly on residents outside the transaction. Our first contribution is to value these broader local effects and study how markets perform when affected communities have the power to block reallocation. By measuring both communities’ value of retaining a resource and the private gains from moving it, we quantify when community veto rights prevent profitable trades and when bargaining can preserve them. This provides an empirical analogue to the classic distinction between property rights and uniform corrective instruments (Coase, 1960, Pigou, 1920).

Research on water markets has established substantial gains from reallocating water toward higher-value uses and documented the institutions that facilitate or impede this reallocation (Young,

1986, Hanemann, 2005, Regnacq et al., 2016, Rafey, 2023, Hagerty, 2023, Ferguson, 2025). Yet much less is known about the consequences for communities from which water leaves or the value residents place on avoiding them, despite these concerns being central to water policy.³ We fill this gap by quantifying these local consequences and community values, bringing a central source of opposition to water markets into the analysis of their gains.

Finally, spatial equilibrium and revealed-preference methods use households’ location choices to value local amenities and changes in place characteristics (Bayer et al., 2007, Diamond, 2016, Almagro & Dominguez-Iino, 2025, Kashner, 2026). We extend these methods to recover heterogeneous welfare effects without requiring separate causal variation for each location. This allows us to characterize the distribution of local welfare effects, including those operating through unobserved amenities, rather than restricting the analysis to an average effect.

The paper proceeds as follows. Section 2 describes the institutional setting and introduces our analytical framework. Section 3 estimates causal effects of persistent reform-induced water reallocation on local economic outcomes. Section 4 introduces our dynamic spatial equilibrium model and Section 5 describes how its estimated. Section 6 compares community and private values for water and discusses the implications for water market design. Section 7 concludes.

2 Framework

Market integration allows property-right holders to realize gains from trade by reallocating assets and resources toward higher value users. However, the effects of trade can extend beyond the transacting parties. The transfer of productive assets may shift economic activity across communities, development can affect nearby residents, and changes in land use can alter local amenities. When these effects fall on parties outside the transaction, who holds the right to allow trade can shape the extent of market reallocation.

Realizable Trades. Consider a seller who holds a transferable good and a buyer who would like to acquire it. Let V_s^P and V_b^P denote their respective private values. The private gains from trade are $G_{sb}^P = V_b^P - V_s^P$, so reallocation creates private surplus whenever $G_{sb}^P > 0$. In our application, the good is a property right to water and these private values are the agricultural profits of the selling and buying farmers.

The transaction may also affect people who do not participate in it, who we refer to as the *community*. When the transaction shifts economic activity across places, let V_s^C denote the selling community’s aggregate willingness to pay to retain the asset locally and V_b^C the buying community’s aggregate willingness to pay for the local benefits of receiving it.⁴

In the free market benchmark, the buyer and seller hold the rights relevant to the transaction.

³Ge et al. (2023) estimates local impacts in a case study of a particular water trade, while Sum (2025) analyzes hypothetical local impacts of groundwater reallocation.

⁴When a transaction changes ownership or use without reallocating the asset across communities, there is instead a single affected community with willingness to pay, V_s^C , for the status quo use.

They can trade whenever $G_{sb}^P > 0$ without obtaining the consent of affected communities. If instead, the selling community is able to block a transfer, the buyer and seller can use their gains from trade to secure their consent through compensation (Coase, 1960). If the selling community's value is known, the traders can fully compensate them from their private gains whenever $G_{sb}^P \geq V_s^C$.

A reallocation that imposes losses on one community may create gains for another. If the transfer benefits residents of the buying community, their value V_b^C provides an additional source of surplus that may be used to purchase the selling community's consent.⁵ If this surplus can be accessed, trades may still proceed whenever $G_{sb}^P + V_b^C \geq V_s^C$.

Discussion. Heterogeneity in community values makes their spatial relationship with private gains from reallocation central to the effects of local control. If water leaving high-value communities generates large private gains elsewhere, those gains may be sufficient to compensate the communities and preserve trade. If high community values instead coincide with small gains from reallocation, community control will block more otherwise profitable trades.

The consequences of community veto rights also depend on information about community values. Under full information, communities can be paid exactly their value. When this value, V_s^C , is private information to the community, traders cannot distinguish communities that require high compensation from those willing to accept less. Higher offers make consent more likely but leave less of the gains from trade to the transacting parties. This information problem allows some communities to capture rents and can prevent trades that could fully compensate the community under full information (Myerson & Satterthwaite, 1983).

Alternatively, a market designer could require compensation without giving affected communities the right to block transactions. Governments could consider a uniform compensation policy under which each selling community receives a posted price when a transfer occurs. Such a policy does not require eliciting each community's value as a condition for trade, but it does not ensure that communities are fully compensated. A payment sufficient to compensate communities with high V_s^C overcompensates those with lower values, while a lower payment leaves some communities undercompensated.

The market consequences of community control therefore depend on the magnitude and heterogeneity of community values, their relationship with the gains from reallocation, the information available to target compensation, and the sources of surplus that can be mobilized to secure consent. The empirical analysis that follows measures these values and their spatial relationship; Section 8 then uses them to quantify how alternative forms of community control affect market reallocation.

⁵Governments frequently mobilize anticipated local gains to influence the location of economic activity. State and local governments, for example, offer firms tax incentives and direct subsidies in exchange for investments and job creation (Slattery, 2025); more broadly, place-based policies use public resources to attract or retain economic activity because of its anticipated benefits to local residents (Kline & Moretti, 2014).

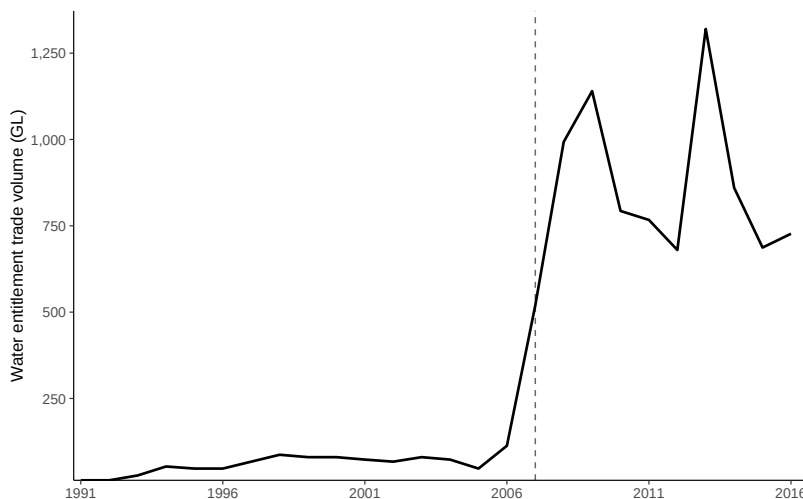
3 Institutional Background and Data

3.1 Water in Murray-Darling Basin

The Murray-Darling Basin (MDB) is Australia’s largest and most economically significant river system, supporting around 2 million residents. It is the same size as Texas and California combined, and is twice as large as France. In Appendix Figure 8, we present a map of the MDB which intersects with Queensland, New South Wales, Victoria, Canberra, and South Australia.

This rural agricultural region is responsible for 40% of Australia’s production in irrigated agriculture. The critical agricultural input is surface water, which is highly variable with annual inflows fluctuating by more than a factor of ten across years (Grafton & Wheeler, 2018). The region’s aridity, volatility, and agricultural dependence has made water management policy a central and contested policy issue. For much of Australia’s modern history, water access was managed by a fragmented system of licenses and water rights that allowed farmers to extract water from local rivers, canals, and irrigation districts. These rights were often tied to land and historical use. In the late 20th century,⁶ while infrastructural and legal advancement allowed for larger-scale water management, water rights remained incredibly heterogeneous, incompatible across state boundaries, and rarely traded in formal markets.

Figure 1: Water entitlement trade volumes over sample period



Notes: Figure reports the annual volume of water entitlement trades in the MDB. Entitlements are permanent rights to a share of available water. Data are from ABARES (Goesch et al., 2019). The dashed line denotes 2007, the beginning of the post-reform period. One GL equals one billion litres, or 1,000 megalitres (approximately 811 acre-feet). For scale, approximately 10,000-12,000 GL are delivered in an average year in the MDB.

These limitations became increasingly salient during the Millennium Drought (1997–2009), one of the worst droughts in Australian history. As inflows reached record lows and irrigation allocations fell to near zero in many regions, political support grew for Basin-wide reform. Beginning with

⁶In 1994, the Council of Australian Governments (COAG) adopted a national water reform framework that promoted the separation of water rights from land which was a key foundation for future water marketing.

the Water Act of 2007, a series of reforms created the most active water market in history by standardizing water rights and trading rules and integrating the market across the Basin. An important component of this integration was limiting local institutions’ ability to block water from leaving their jurisdictions. New water-market rules restricted irrigation districts’ ability to impede out-of-district transfers and capped the exit fees they could charge departing entitlement holders. Market integration therefore involved not only making water rights more tradable, but also curtailing local control over whether water could leave a community. Figure 1 shows the resulting rise in water-right transfers, with post-reform trade volumes reaching nearly twenty times their pre-reform levels. The reforms also facilitated large environmental water buybacks, contributing to a substantial spatial reallocation of irrigation activity across the Basin.

Proponents of the reforms emphasized the efficiency and sustainability benefits of water trade. [Rafey \(2023\)](#) finds that post-reform allocations shifted water toward more productive agricultural users, creating value equivalent to 8–12% of total annual supply. The movement of water out of some regions, however, generated concerns about the consequences for selling communities. These concerns fueled protests and political opposition to the reforms and generated calls to restore restrictions on trade or compensate affected communities. The Australian experience re-affirmed skepticism in other parts of the world about market reallocation of water and illustrates the central tension of this paper: removing local barriers can facilitate productive reallocation while creating local consequences that generate demand for greater control over that reallocation.

3.2 Data

We combine satellite measures of water use with administrative and Census data on households, local economic activity, migration, and agricultural production. Our primary unit of analysis is the Statistical Area Level 1 (SA1), the smallest standard geography for which the Australian Census releases detailed population data. SA1s generally contain between 200 and 800 residents and average roughly 400 residents.⁷ Our Murray–Darling Basin sample contains 6,494 SA1s nested within 383 Statistical Area Level 2 (SA2) regions, which we use when finer information is unavailable.

3.2.1 Water use and agricultural conditions

Evapotranspiration. Our primary measure of local water use is actual evapotranspiration (ET) from the Terrestrial Ecosystem Research Network’s CMRSET product from 2000–2020. The data provide monthly estimates at 30-meter resolution across Australia. CMRSET combines remotely sensed vegetation information from high-resolution satellites with higher-frequency satellite observations and gridded meteorological data to produce continuous monthly estimates of actual evapotranspiration. We average pixel-level estimates within each SA1 and construct alternative measures that exclude pixels within 100, 250, and 500 meters of major surface waterways. Our preferred specification excludes pixels within 500 meters of waterways.

⁷SA1s are roughly comparable in scale to U.S. Census block groups.

Climate. We supplement ET with gridded rainfall and potential evapotranspiration from the Australian Bureau of Meteorology from 2000–2020. These variables capture local meteorological conditions that generate evapotranspiration in the absence of irrigation.

Administrative water data. We collect annual administrative data on water allocations and trade from the Murray–Darling Basin Authority from 2000–2020. These data are reported at water-management geographies substantially larger than an SA1. We use them to validate our satellite-based measure of irrigation activity.

Soil. We obtain predetermined soil characteristics from the Soil and Landscape Grid of Australia. The Grid combines soil observations, laboratory measurements, remote sensing, and digital-soil-mapping methods to predict non-anthropogenic soil attributes on an approximately 90-meter grid. We use measures of soil depth and available water capacity to construct instruments for post-reform changes in irrigation.

3.2.2 Households and local economic outcomes

Census outcomes. The Australian Census provides SA1-level measures of population, household income, rents, mortgage payments, employment, and residential tenure every five years from 2001 through 2021. We supplement these data with archived 1991 and 1996 Census geography files obtained through the Australian Urban Research Infrastructure Network (AURIN) when available. Appendix Table 6 reports summary statistics for key Census variables.

Migration and tenure. For the 2006, 2011, 2016, and 2021 Censuses, the ABS reports individuals’ current residence together with their residence one and five years earlier. We observe these migration histories by age group and use them to construct annual migration choices, household tenure states, and the local stock of long-term residents used in the residential-choice model.

Agricultural production. The ABS reports agricultural production at the SA2 level at approximately five-year intervals between 2005 and 2020. For each crop, we observe the value of agricultural output, cultivated land, and applied irrigation water. We combine these data with agricultural employment and wages to adapt the production model of [Rafey \(2023\)](#) to our geographic units and recover location-specific marginal productive values of water. For estimation, we aggregate crops into three categories: annual crops, perennial crops, and dairy/pasture.

Additional local outcomes. We supplement the Census with SA2-level data on business activity, mortality, suicide, and federal-election outcomes. These outcomes allow us to examine whether the local effects of water reallocation extend beyond population, income, and housing markets.

4 Local Consequences of Water Reallocation

We first examine how market integration changed the spatial allocation of irrigation and the consequences of this reallocation for affected communities. We show that the reforms generated substantial and persistent shifts in irrigation activity across the Basin. Predetermined differences in agricultural suitability strongly predict the direction of these shifts, providing differential exposure to reform-induced water reallocation. We use this variation to estimate the effects of water gains and losses on local economic and social outcomes. Later in the paper, we use the same source of variation to estimate households’ overall valuation of local irrigation.

4.1 Water Reallocation Following Market Integration

Measuring local irrigation. While Figure 1 shows the sharp increase in water trading after the reform, it does not reveal the resulting spatial reallocation of water. Because granular administrative data on irrigation are unavailable, we construct a local measure of irrigation activity using satellite evapotranspiration (ET) and weather data. Our preferred measure residualizes monthly ET per hectare against current and lagged rainfall, potential evapotranspiration, and SA1-by-calendar-month fixed effects, excluding pixels within 500 meters of major waterways. We define annual local irrigation, W_{jt} , as the sum of residual ET over the October–March dry season, expressed in ML per hectare. We validate this measure against administrative water data: an additional ML of water allocation is associated with approximately 0.65 ML of dry-season residual ET, while allocations have essentially no relationship with wet-season residual ET.⁸ Appendix Section A describes the construction and validation exercises in detail.

We measure a community’s exposure to water reallocation as the change in average irrigation activity since the reform relative to its pre-reform average,

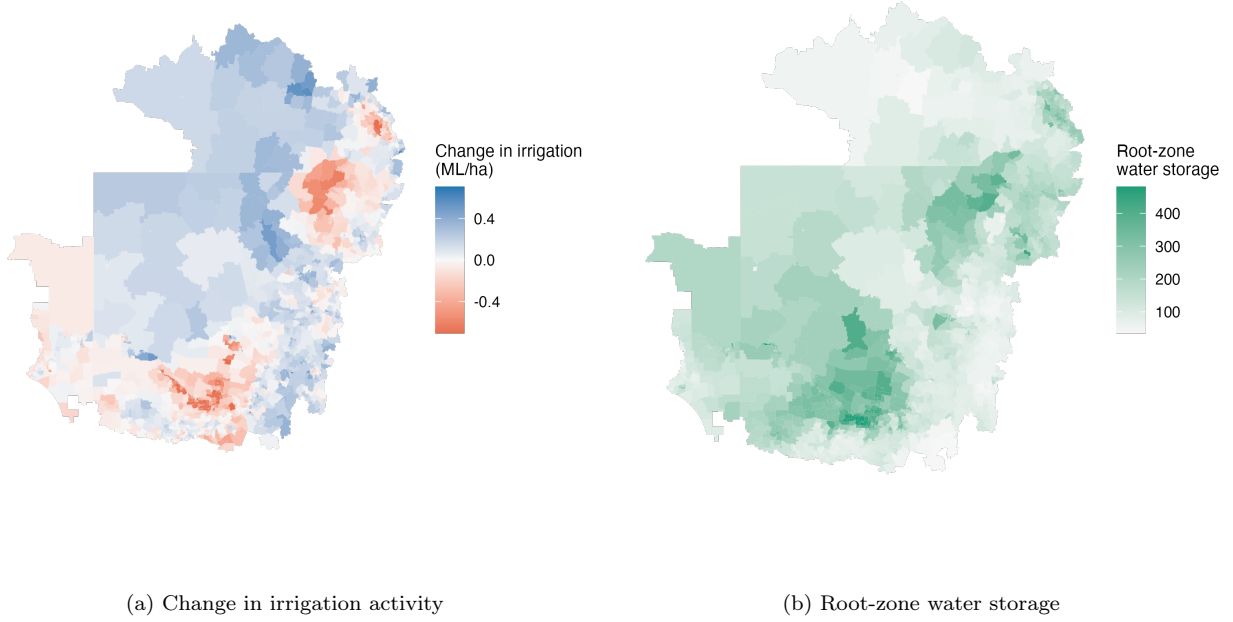
$$\Delta W_{j,t} = \overline{W}_{j,\text{post}\leq t} - \overline{W}_{j,\text{pre}}, \quad (1)$$

allowing exposure to evolve as water is progressively reallocated across the Basin. Panel (a) of Figure 2 maps this measure for $t = 2021$ which corresponds to the average post-reform change in irrigation activity relative to the pre-reform period. The median community experienced an absolute change of 0.15 ML/ha, equivalent to approximately 16% of applied irrigation; at the 10th and 90th percentiles, changes were approximately -37% and 26% , respectively. The median change was modestly negative, consistent with the substantial volume of water purchased through the government’s environmental water buyback program.

Differential exposure to reform. The substantial gains and losses in irrigation documented above provide differential exposure to market integration across communities. To predict this reallocation, we use root-zone water storage, a predetermined soil characteristic constructed as the

⁸This magnitude is consistent with estimates of irrigation efficiency in the MDB: Roth et al. (2013) report whole-farm irrigation efficiencies of approximately 57–70%.

Figure 2: Water Reallocation and Predetermined Agricultural Suitability



Notes: Panel (a) maps the change in local irrigation activity between the pre-reform period and 2021, measured in ML per hectare. Panel (b) maps root-zone water storage, constructed as the interaction of available water capacity and soil depth. Color scales are capped at the 1st and 99th percentiles to limit the influence of extreme values.

interaction of available water capacity and soil depth. Root-zone water storage captures the amount of water that can be stored within the soil profile and made available to plant roots. Soils with lower root-zone storage may still be productive, but depend more heavily on reliable irrigation supply. Panel (b) of Figure 2 maps root-zone water storage across the Basin. Comparing Panels (a) and (b) reveals a clear spatial relationship: areas with lower root-zone water storage tend to experience larger increases in irrigation following market integration.

We quantify this relationship by estimating

$$\Delta W_{j,t} = \pi_t Z_j + \gamma_{d(j)} + \nu_{j,t}, \quad (2)$$

where Z_j is root-zone water storage and $\gamma_{d(j)}$ denotes water-region fixed effects. Root-zone water storage negatively predicts changes in irrigation in each post-reform period, consistent with irrigation expanding relatively more on soils that depend more heavily on reliable water supply. The relationship is strong, with F -statistics ranging from 28.0 to 35.6 across post-reform years. Appendix Table 7 reports the estimates. The interaction of this predetermined soil variation with market integration therefore provides plausibly exogenous variation in local irrigation, which we use below to estimate the effects of water gains and losses on affected communities.

4.2 Local Economic Consequences

We next examine how the reallocation of water across the Basin affects the communities that gain and lose irrigation. We use the differential exposure documented above to estimate these local effects. For outcome Y_{jt} , we estimate

$$\Delta Y_{j,t} = \beta_t \Delta W_{j,t} + \gamma_{d(j)} + \varepsilon_{j,t}, \quad (3)$$

where $\Delta Y_{j,t}$ is the change in the outcome relative to its pre-reform baseline and $\Delta W_{j,t}$ is the corresponding change in local irrigation. We instrument $\Delta W_{j,t}$ using the predetermined soil variation described in Equation 2. In the post-reform period, water exposure is measured through the corresponding outcome year, allowing the treatment to evolve as irrigation is progressively reallocated across the Basin. Standard errors are clustered at the SA2 level.

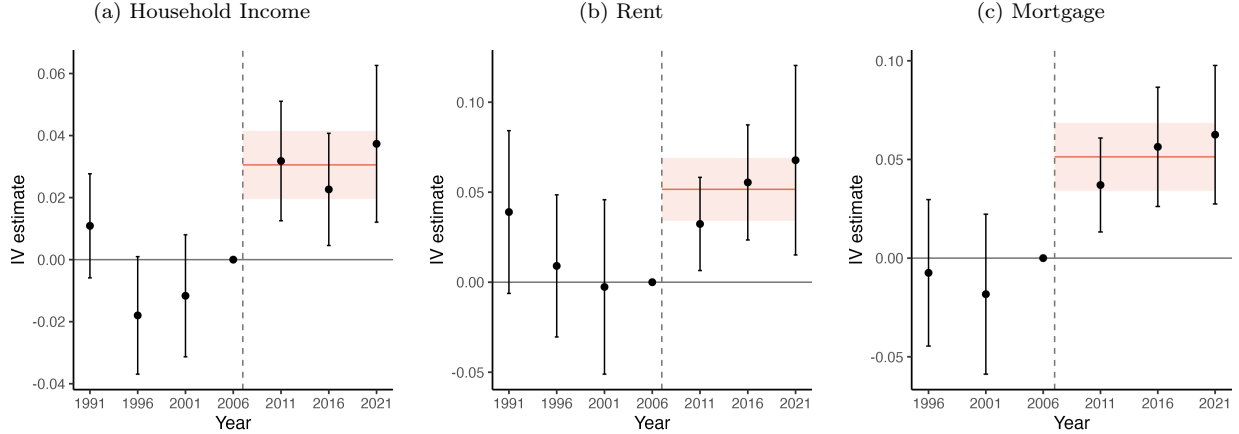
For outcomes observed before market integration, we use the same design to test whether communities that would subsequently experience different water changes were already on different trajectories (Alsan et al., 2025). In these pre-reform specifications, we use eventual post-reform water reallocation as the treatment and instrument it with the same predetermined soil variation. Identification requires that this variation affect changes in local outcomes following market integration through its effect on irrigation rather than through other differential changes across communities. The pre-reform estimates therefore provide a direct test for differential trends associated with our instrument.

Local economic activity. Figure 3 reports effects on the three economic outcomes that we observe both before and after the reform: household income, rent, and mortgage payments. We find immediate and persistent effects on all three. A 10% reduction in local irrigation reduces household income by approximately 3.0% and housing costs by approximately 5.1%, with similar estimates across post-reform years. The pooled estimates summarize these persistent effects. The longer panels for these outcomes also allow us to evaluate pre-trends. We find no evidence that communities subsequently exposed to water gains or losses were already experiencing differential trends in income or housing costs before the reform.

For outcomes available only after the reform, we estimate pooled specifications using the same instrument-induced variation in irrigation. Appendix Table 8 reports the pooled estimates for all outcomes discussed in this subsection. A 10% reduction in irrigation reduces agricultural employment by approximately 7.5% and the number of operating businesses by 2.6%. We estimate a precise near zero effect on personal incomes, so the reform-induced change in irrigation seems to affect household incomes through employment, rather than through wages.

Household migration and social response. Households respond to these changes in local economic conditions. A 10% reduction in irrigation reduces population by 2.7% in 2011 and 7.1% by 2021, with a pooled effect of 4.5%. We find a similarly gradual response in the stock of long-term residents, a measure of the persistence of the local community: the effect grows from 1.3% in 2011

Figure 3: Effect of Water Gain on Income, Rent, and Mortgage



Notes: Figure depicts the effect of a persistent increase of 0.096 ET/hectare on SA1 incomes, rents, and mortgage payments relative to 2006 for each Census year from 1996-2021. We instrument for water change with soil characteristics. One unit of water change is equivalent to a 10% increase in irrigation intensity at the average SA1. Annual estimates of the effect are reported in black dots and the pooled post-reform effect is the horizontal red line. Standard errors are clustered at the SA2 level for annual results and at the SA1 level for the pooled effect.

to 6.9% by 2021, with a pooled effect of 3.7%. We depict these transitions in Appendix Figure 9. Evidence of this gradual response relative to local economic indicators motivates migration frictions and a dynamic response to reform for our residential choice model in Section 5.

Water losses also produce political and social responses. A 10% reduction in irrigation increases the Labour vote share by approximately 2.1 percentage points and deaths of despair by 17.4%. These results reinforce that water reallocation affects communities along dimensions beyond the economic outcomes observed above.

4.3 From Local Consequences to Community Values

The results above show that water reallocation affects communities along multiple margins, from income and housing costs to migration, political behavior, and social distress. Many of these consequences cannot be directly expressed in dollars, yet the framework in Section 2 requires an encompassing measure of residents' willingness to pay to retain water locally. At the same time, the reduced-form results provide support for our empirical design: predetermined soil variation strongly predicts post-reform water reallocation, but does not predict differential trends in the outcomes we observe before the reform. We therefore use this same variation in a dynamic residential choice model, recovering community willingness to pay from households' decisions to remain in, leave, or move to communities as local irrigation changes.

5 Model

We recover the value residents place on retaining the local benefits of irrigation from their revealed residential choices. Households reveal their preferences over communities through decisions to re-

main, leave, or move into them. Because these decisions are persistent and forward-looking, we model them using a dynamic residential choice framework. The model allows water to affect the attractiveness of a community through observed economic and demographic conditions as well as unobserved local attributes. We use the resulting flow utilities to define the community value of water. We then define the productive value of water from agricultural producers' profit maximization, providing the private-value benchmark for our market-design analysis.

5.1 Residential Choice

There are $J + 1$ locations: J SA1s in the Murray-Darling Basin, indexed by $j \in \mathcal{J}$, and an outside option, $j = 0$, representing the rest of Australia. At the beginning of each period, household h chooses where to live. Its individual state is

$$x_{ht} = \{j_{h,t-1}, \tau_{h,t-1}\}, \quad (4)$$

where $j_{h,t-1}$ is the household's previous location and $\tau_{h,t-1}$ is its tenure in that location.

Flow utility. The flow utility household h receives from choosing location j is

$$u_{ht}(j) = \bar{u}_{jt\tau} + MC_t(j, j_{h,t-1}) + \varepsilon_{hjt}, \quad (5)$$

where τ denotes the household's tenure state, $\bar{u}_{jt\tau}$ is flow indirect utility from residing in location j , $MC_t(\cdot)$ captures moving costs, and ε_{hjt} is an independently and identically distributed Type-I extreme-value preference shock.

We parameterize flow indirect utility as

$$\bar{u}_{jt\tau} = \alpha_I \log I_{jt} + \alpha_p \log p_{jt} + \alpha_W W_{jt} + \alpha_T T_{jt} + \lambda_\tau + \xi_{jt\tau}, \quad (6)$$

where I_{jt} is household income, p_{jt} is the rental price, W_{jt} is local irrigation, and T_{jt} is the stock of long-term residents. The tenure-state fixed effects λ_τ capture differences in utility associated with a household's own tenure in a location, while α_T captures households' preference for living in communities with a larger stock of long-term residents. The term $\xi_{jt\tau}$ captures other unobserved determinants of local utility.

This specification distinguishes a household's own attachment to place from its preference for the persistence of the surrounding community. It also does not require us to enumerate every consequence of water reallocation. Water may affect utility directly, through income and the persistence of the local community, or through other valued attributes of place captured by $\xi_{jt\tau}$. The community value we recover below incorporates all of these channels.

Dynamic location choice. Residential choices exhibit substantial persistence: households move infrequently, and the probability of leaving a community varies systematically with how long a household has lived there. We capture these features through moving costs and tenure-dependent

flow utility. Because today's residential choice determines both the household's future location and its tenure state, households are forward-looking.

The household's value function satisfies

$$V_t(x_{ht}, \varepsilon_{ht}) = \max_j \{u_{ht}(j) + \beta \mathbb{E}[V_{t+1}(x_{h,t+1}, \varepsilon_{h,t+1}) \mid j, x_{ht}]\}, \quad (7)$$

where $\beta \in (0, 1)$ is the discount factor. A household that remains in its current location preserves and accumulates tenure, while a household that moves begins with no tenure in its new location.

The dynamic structure is important for recovering flow utility from residential choices. An observed decision to remain in a community reflects not only the utility the household receives today, but also moving costs and the consequences of the decision for future location and tenure. Section 6.1 shows how observed migration choices and a renewal structure in the household state allow us to recover $\bar{u}_{jt\tau}$ while accounting for these forward-looking incentives.

5.1.1 Community Value of Water

We use the residential choice model to measure the value households place on the consequences of local water reallocation. Irrigation can affect household utility both through its conditional effect in Equation (6) and by changing other attributes of the community. The total effect of irrigation on household flow utility is

$$\frac{d\bar{u}_{jt}}{dW_{jt}} = \underbrace{\frac{\partial \bar{u}_{jt}}{\partial W_{jt}}}_{\text{conditional water effect}} + \underbrace{\frac{\partial \bar{u}_{jt}}{\partial \log I_{jt}} \frac{d \log I_{jt}}{dW_{jt}} + \frac{\partial \bar{u}_{jt}}{\partial \log p_{jt}} \frac{d \log p_{jt}}{dW_{jt}} + \frac{\partial \bar{u}_{jt}}{\partial T_{jt}} \frac{dT_{jt}}{dW_{jt}}}_{\text{observed equilibrium effects}} + \underbrace{\frac{d\xi_{jt}}{dW_{jt}}}_{\text{other local attributes}}. \quad (8)$$

These responses may differ across communities. To make the sources of this heterogeneity explicit, write

$$\frac{d \log I_{jt}}{dW_{jt}} = \beta_I + \eta_j^I, \quad \frac{dT_{jt}}{dW_{jt}} = \beta_T + \eta_j^T, \quad \frac{d\xi_{jt}}{dW_{jt}} = \eta_j^W, \quad (9)$$

where the η_j terms capture community-specific departures from the average response. We normalize each heterogeneous component to have mean zero across communities.

We assume that changes in local housing costs affect residential choices, but we treat changes in housing expenditures induced by water reallocation as transfers between residents and local property owners rather than as a net cost to the community.⁹

We therefore define non-housing flow utility as

$$\zeta_{jt\tau} \equiv \bar{u}_{jt\tau} - \alpha_p \log p_{jt} - \lambda_\tau. \quad (10)$$

⁹This treatment assumes that changes in housing expenditures induced by water reallocation accrue to property owners within the community. We nevertheless retain housing prices in household utility because they affect residential choices and, below, provide the numeraire used to express utility values in monetary terms.

The effect of irrigation on this utility index, $\frac{d\zeta_{jt}}{dW_{jt}}$, contains each of the channels in Equations 8 other than the housing-transfer component. We denote this encompassing effect of irrigation on community utility by ω_j which we summarize with a mean and location specific components as

$$\omega_j \equiv \frac{d\zeta_{jt}}{dW_{jt}} = \bar{\omega} + \eta_j, \quad \mathbb{E}[\eta_j] = 0. \quad (11)$$

The component $\bar{\omega}$ is the mean utility effect of irrigation and η_j captures heterogeneity in community values. We do not estimate ω_j or η_j by separately estimating and summing these channels. Instead, our empirical strategy estimates their total effect on local utility.¹⁰ Section 6.1.3 shows how repeated reform-induced changes in water allow us to recover the distribution of this total community value even when location-specific causal effects cannot be estimated individually.

We sum up household ω_j to obtain aggregate community value by

$$V_{jt}^C = M_{jt}\omega_j, \quad (12)$$

where M_{jt} is the number of households in community j . This value is initially expressed in utility units. We use households' estimated housing-price preferences to translate it into dollars, making community values directly comparable to the private value of water.

5.2 Productive Value of Water

To compare community values with the private gains from water reallocation, we also require the marginal value of water to agricultural producers. A producer of crop c in location j produces output using irrigation water W_{jct} , rainfall R_{jct} , cultivated land K_{jct} , labor L_{jct} , materials M_{jct} , and productivity ω_{jct} . Conditional on crop choice and cultivated land, the producer chooses variable inputs to maximize operating profits:

$$\pi_{jct}(W_{jct}) = \max_{L_{jct}, M_{jct}} \{F_{jc}(W_{jct}, R_{jct}, K_{jct}, L_{jct}, M_{jct}; \omega_{jct}) - w_{jt}L_{jct} - q_{jct}M_{jct}\}. \quad (13)$$

Farms are price-takers in the market for commodities, labor, and materials. By the envelope theorem, the marginal effect of water on optimized operating profits is

$$V_{jct}^P \equiv \frac{\partial \pi_{jct}}{\partial W_{jct}} = F_{W,jc}. \quad (14)$$

Thus, the productive value of water is its marginal contribution to farm profits after allowing producers to optimally adjust variable inputs. Section 6.1.3 describes how we estimate $F_{W,jc}$ and aggregate crop-level values to construct the location-level private values used in our comparison with community willingness to pay.

¹⁰Using our causal estimates of water on local outcomes and our demand estimates, we can get a sense of which channels are contributing to the total value of irrigation in a community.

6 Estimation

The framework requires two empirical objects: the value communities place on retaining water locally and the private value of water to agricultural producers. We recover community values from a dynamic residential choice model that relates households’ migration decisions to the flow utility of living in each community. We construct productive values by adapting existing farm-level production estimates to our regional data.

6.1 Residential Choice and Community Water Values

We use households’ observed residential choices to recover the flow utility of living in each community. Our data record flows of households’ previous and current locations as well as their tenure in their previous location. These migration histories allow us to distinguish the utility associated with remaining in a community from moving costs and the forward-looking value of preserving tenure.

We estimate the model using a two-step, BLP-style procedure adapted to our dynamic location-choice setting (Davis et al., 2023, Berry et al., 1995). First, we use conditional choice probabilities and the renewal structure of the dynamic problem to construct a continuation-value correction. Conditional on this correction, maximum likelihood recovers tenure-specific mean flow utilities and moving costs from observed origin-destination flows. Second, we decompose the recovered flow utilities into income, housing prices, irrigation, community tenure stock, and unobserved local amenities using instrumental variables.

6.1.1 Recovering Flow Indirect Utilities

We first derive an equation relating flow indirect utility and moving costs to observed conditional choice probabilities. Closely following Almagro & Dominguez-Iino (2025), we use a Hotz–Miller inversion together with restrictions on the evolution of household tenure. The resulting renewal structure eliminates otherwise unknown continuation values from the estimating equation. We provide the full derivation in Appendix Section B.

We begin with standard assumptions on the dynamic discrete-choice environment.

Assumption 1 (Dynamic choice environment). *Households are atomistic and the household state follows a Markov process with a conditional-independence structure. In particular, an individual household’s residential choice does not affect the evolution of the aggregate state,*

$$\mathbb{P}(\Omega_{t+1} \mid j, x_t, \Omega_t, \varepsilon_t) = \mathbb{P}(\Omega_{t+1} \mid \Omega_t), \quad (15)$$

while the evolution of the household-specific state depends only on the current state, aggregate state, and residential choice. Idiosyncratic preference shocks ε_{hjt} are independently and identically distributed Type-I extreme value across households, locations, and periods.

Assumption 1 gives the household problem the standard conditional-independence structure of a dynamic discrete-choice model. Current idiosyncratic preference shocks do not predict future states

or preference shocks, and an individual household takes the evolution of aggregate conditions as given. Going forward, we leave dependence on the aggregate state implicit. We normalize the flow utility of the outside option to zero.

Let $\bar{V}_t(x)$ denote the expected value of entering period t in household state x , prior to the realization of idiosyncratic preference shocks. Iteratively applying the Hotz–Miller inversion, the log probability of a household with state $x_{t-1} = (j_{t-1}, \tau_{t-1})$ choosing location j relative to the outside option j_0 can be written as

$$\begin{aligned} \log \frac{\mathbb{P}_t(j \mid x_{t-1})}{\mathbb{P}_t(j_0 \mid x_{t-1})} &= \bar{u}_{jt\tau_t(j, \tau_{t-1})} - \bar{u}_{j_0t,0} + MC_t(j, j_{t-1}) - MC_t(j_0, j_{t-1}) \\ &\quad + \beta [MC_{t+1}(j_0, j) - MC_{t+1}(j_0, j_0)] \\ &\quad - \beta \log \frac{\mathbb{P}_{t+1}(j_0 \mid j, \tau_t(j, \tau_{t-1}))}{\mathbb{P}_{t+1}(j_0 \mid j_0, 0)} \\ &\quad + \beta^2 [\bar{V}_{t+2}(j_0, \tau_{t+1}(j_0, \tau_t(j, \tau_{t-1}))) - \bar{V}_{t+2}(j_0, 0)]. \end{aligned} \quad (16)$$

Equation 16 separates current flow utility and moving costs from the forward-looking consequences of the residential choice. Without further restrictions, estimation would still require solving for the continuation values in the final line. We instead exploit the structure of tenure and the composite outside option to construct a renewal path under which the two choice histories reach the same payoff-relevant state after two periods.

Renewal through the outside option. We model tenure as a payoff-relevant state only while a household resides within the Murray-Darling Basin. A household that moves to a new MDB location begins with no tenure in that location, while remaining in the same location preserves and increments tenure. Choosing the composite outside option resets modeled tenure.

Assumption 2 (Tenure and outside-state renewal). *Household tenure evolves according to*

$$\tau_t(j, \tau_{t-1}) = \begin{cases} \tau_{t-1} + 1, & j = j_{t-1}, \quad j \neq j_0, \\ 0, & j \neq j_{t-1}, \end{cases} \quad (17)$$

and tenure is not modeled within the composite outside option, $\tau_t(j_0, \tau) = 0$ for all τ .

Assumption 2 makes the outside option a renewal state. Consider a household that chooses location j in period t and the outside option in period $t + 1$, relative to a household choosing the outside option in both periods. Both histories arrive in period $t + 2$ in the same payoff-relevant state, $(j_0, 0)$. Hence,

$$\bar{V}_{t+2}(j_0, 0) - \bar{V}_{t+2}(j_0, 0) = 0, \quad (18)$$

and all continuation values from period $t + 2$ onward cancel. This finite-dependence property allows us to estimate the dynamic choice model without explicitly computing household value functions.

We parameterize contemporaneous moving costs as a linear function of distance,

$$MC_t(j, j_{t-1}) = \kappa \text{Dist}(j, j_{t-1}). \quad (19)$$

The renewal expression also generates moving-cost components associated with subsequently reaching the outside option. Components that vary only with the household's origin are absorbed by origin fixed effects, while destination-specific components are absorbed into the destination flow utilities recovered below. Appendix Section B provides the complete decomposition.

Observed tenure states and migration histories. For each Census year $t \in \{2006, 2011, 2016, 2021\}$, we observe a household's location five years prior, j_{t-5} , one year prior, j_{t-1} , and at the time of the Census, j_t . We treat the transition from j_{t-1} to j_t as the annual residential choice and use j_{t-5} to construct the household's tenure state.

Because locations at $t - 5$ and $t - 1$ are observed at the SA2 level, tenure is defined with respect to residence in the same SA2.¹¹ Immediately prior to the annual choice,

$$\tau_{t-1} = \begin{cases} 4+, & j_{t-5} = j_{t-1}, \\ [0, 4), & j_{t-5} \neq j_{t-1}. \end{cases} \quad (20)$$

Combining this state with the observed annual choice gives

$$\tau_t(j, \tau_{t-1}) = \begin{cases} 5+, & j = j_{t-1} \text{ and } \tau_{t-1} = 4+, \\ [1, 5), & j = j_{t-1} \text{ and } \tau_{t-1} = [0, 4), \\ [0, 1), & j \neq j_{t-1}. \end{cases} \quad (21)$$

These states capture the strong state dependence in residential mobility observed in the data: long-term residents are substantially less likely to leave their communities than recent arrivals. Together with geographic moving costs, this pattern motivates the dynamic structure of the model. An observed decision to remain therefore cannot be interpreted solely as a preference for the community's current attributes; it also reflects the costs of moving and the continuation value associated with the household's tenure state.

These migration histories identify current annual choice probabilities conditional on tenure. The continuation correction additionally requires exit probabilities in the subsequent annual period, which we do not observe directly. We therefore impose a local stationarity approximation.

Assumption 3 (Stationary-policy approximation). *The annual exit policy is locally stable between adjacent periods. For any MDB location j and tenure state τ ,*

$$\mathbb{P}_{t+1}(j_0 \mid j, \tau) \approx \mathbb{P}_t(j_0 \mid j, \tau). \quad (22)$$

¹¹This allows households to maintain tenure within regions with an average population of approximately 10,000 even if they change residences within the region.

Let $\hat{q}_t(j_0 | x)$ denote the estimated annual exit probability. We summarize the forward-looking component of the location-choice problem with the continuation correction

$$\mathcal{C}_t(j, x_{t-1}; \hat{q}) \equiv \beta \log \frac{\hat{q}_t(j_0 | j, \tau_t(j, \tau_{t-1}))}{\hat{q}_t(j_0 | j_0, 0)}. \quad (23)$$

The assumptions above reduce the dynamic location-choice problem to a standard conditional-choice index with an observed continuation correction. For each inside location $j \neq j_0$, define

$$G_{jt}(x_{t-1}; \theta, \hat{q}) = \delta_{jt\tau} + \gamma_{jt-1} + \kappa \text{Dist}(j, j_{t-1}) - \mathcal{C}_t(j, x_{t-1}; \hat{q}), \quad (24)$$

where $\delta_{jt\tau}$ is the flow indirect utility associated with destination j , period t , and tenure state τ , γ_{jt-1} captures origin-specific components of moving costs, and κ governs the effect of distance on moving costs.

Conditional on the estimated continuation correction, choice probabilities take the familiar logit form,

$$P_t(j | x_{t-1}; \theta, \hat{q}) = \frac{\exp\{G_{jt}(x_{t-1}; \theta, \hat{q})\}}{1 + \sum_{\ell \neq j_0} \exp\{G_{\ell t}(x_{t-1}; \theta, \hat{q})\}}, \quad j \neq j_0. \quad (25)$$

Let $N_t(j; x)$ denote the number of households observed in state x that choose location j . We estimate the first-stage parameters by maximum likelihood,

$$\hat{\theta} = \arg \max_{\theta} \sum_{t,x} \sum_{j \in \mathcal{J}} N_t(j; x) \log P_t(j | x; \theta, \hat{q}). \quad (26)$$

The first step therefore uses observed migration choices to recover tenure-specific flow indirect utilities $\delta_{jt\tau}$ while accounting for moving costs and the forward-looking consequences of current residential choices. Importantly, we do not need to solve the household's value function or specify a parametric law of motion for the aggregate state. The dynamic component is summarized by the continuation correction in Equation 23.

The MLE formulation also substantially reduces the need to smooth empirical conditional choice probabilities. Many destination choices are never observed from a particular location-tenure state, but zero choice counts simply contribute zero to the likelihood. Regularization is required only for the exit probabilities entering Equation 23, since these appear inside a logarithm and must be strictly positive. We regularize these probabilities with a Beta-Binomial estimator that adds a small number of prior pseudo-counts to observed outside and inside choices. Appendix Section B.4 provides details.

6.1.2 Location Preferences

In the second step, we decompose the recovered flow indirect utilities into the location characteristics introduced in Section 5. Specifically,

$$\hat{\delta}_{jt\tau} = \alpha_I \log I_{jt} + \alpha_p \log p_{jt} + \alpha_W W_{jt} + \alpha_T T_{jt} + \lambda_{\tau} + \gamma_j + \gamma_{r(j),t} + \xi_{jt\tau}, \quad (27)$$

where λ_τ are tenure-state fixed effects, γ_j are location fixed effects, and $\gamma_{r(j),t}$ are water-region-by-year fixed effects. Because income, housing prices, water use, and tenure stock may respond to unobserved changes in local amenities, we instrument for each. The water instrument is discussed in Section 4 and we discuss the construction of the tenure stock, income, and price instruments in Appendix Section C.

Water. We instrument for irrigation W_{jt} using the same predetermined soil variation interacted with water-market reform timing used in Section 4. The instrument therefore isolates reform-induced changes in local irrigation rather than variation arising from contemporaneous local conditions.

Tenure stock. Unobserved amenities may both increase residential persistence and make a location more attractive. We instrument for T_{jt} using a demographic shift-share instrument that interacts each SA1’s lagged age-by-tenure composition with age-specific residential retention rates calculated outside the SA1’s own SA2. The instrument predicts changes in tenure stock from predetermined demographic composition and the propensity of similar households elsewhere to remain in place.

Income. We instrument for household income using a Bartik-style shift-share instrument that interacts baseline industry employment shares with national industry-level employment growth. Identification comes from differential exposure to national employment changes induced by pre-existing local industry composition.

Housing prices. We instrument for housing prices using predicted local mortality shocks, following Kashner (2026). We interact lagged SA1 age shares with national age-specific mortality rates, generating variation in local housing demand from predetermined demographic composition rather than contemporaneous local conditions.

Given instruments

$$Z_{jt} = [z_{jt}^p, z_{jt}^W, z_{jt}^I, z_{jt}^T],$$

and the moment condition $\mathbb{E} [Z_{jt} \xi_{jt\tau}^k] = 0$, we estimate the relevant coefficients via IV-GMM. With multiple endogenous community characteristics, identification additionally requires that the instruments generate sufficiently independent variation to separately identify their effects. Formally, the matrix of first-stage relationships between the instruments and endogenous characteristics must have full column rank.¹²

6.1.3 Community Value of Water

The model defines community water value as the encompassing effect of irrigation on non-housing flow utility. Rather than constructing this value from separate estimates of the channels in Equation

¹²The reported standard errors do not yet account for sampling uncertainty in the first-stage MLE estimates of $\hat{\delta}_{jt\tau}$. In ongoing work we are developing a bootstrap that re-estimates both stages to incorporate this source of uncertainty.

(8), we estimate it directly with the same differential exposure to reform used in Section 4.

We construct the empirical analogue of the non-housing flow utility index, $\widehat{\zeta}_{jt\tau}$ defined in Equation (10) and take differences relative to the pre-reform period to get $\Delta\widehat{\zeta}_{jt\tau}$. Define $\Delta\widehat{\zeta}_{jt}$, which is the tenure state population weighted change in household flow utility net of housing price. Our definition of ω_j from Equation (11) implies

$$\Delta\widehat{\zeta}_{jt} = (\bar{\omega} + \eta_j)\Delta W_{jt} + e_{jt}. \quad (28)$$

where e_{jt} captures changes in local utility unrelated to the persistent community-specific effect of irrigation.

Mean community value. We first estimate the mean effect, $\bar{\omega}$, by applying the same research design used to estimate the local consequences of water reallocation in Section 4, now using $\Delta\widehat{\zeta}_{jt}$ as the outcome. We instrument cumulative changes in irrigation with the reform-induced variation predicted by predetermined soil characteristics. The resulting IV coefficient identifies $\bar{\omega}$, the average effect of irrigation on non-housing flow utility. This procedure estimates the encompassing effect directly and does not require us to separately estimate or monetize each channel through which irrigation affects communities.

Heterogeneity in community value. Community willingness to pay for water also includes the location specific component, η_j . Individual location-specific effects cannot be estimated precisely from the small number of post-reform observations available for each community. Instead, we first recover the dispersion of persistent water effects across communities and then use an Empirical Bayes procedure to recover location-specific values.

Define the residual change in utility, r_{jt} , after removing the mean effect by,

$$r_{jt} \equiv \Delta\widehat{\zeta}_{jt} - \widehat{\bar{\omega}}\Delta W_{jt} = \eta_j\Delta W_{jt} + e_{jt}. \quad (29)$$

Heterogeneity in community water values therefore generates greater dispersion in residual utility changes among communities experiencing larger water reallocations, which we can test for after obtaining our estimates by plotting dispersion in residuals against reform-induced water changes.

A single cross section cannot distinguish persistent heterogeneity in water values from heteroskedastic transitory shocks. We therefore exploit repeated post-reform observations within communities. For two post-reform periods $t \neq s$,

$$\mathbb{E}[r_{jt}r_{js} \mid \Delta W_{jt}, \Delta W_{js}] = a_{ts} + \sigma_\eta^2 \Delta W_{jt} \Delta W_{js}, \quad \sigma_\eta^2 \equiv \text{Var}(\eta_j). \quad (30)$$

The pair-specific intercept a_{ts} allows residual utility changes to remain correlated across periods for reasons unrelated to heterogeneous water effects. Because the persistent component η_j enters the residual in each period multiplied by that period's water exposure, the coefficient on $\Delta W_{jt} \Delta W_{js}$ identifies the variance of persistent community-specific water effects. We estimate σ_η^2 from Equation

30 by pooling the 2011–2016, 2011–2021, and 2016–2021 pairs and using the soil-predicted component of cumulative irrigation changes in constructing the exposure terms.

Location-specific values. The second-moment estimator identifies the dispersion of community values but does not assign a value to each community. We recover location-specific effects using an Empirical Bayes procedure that combines the estimate population distribution of η_j with the three residual utility responses observed for each location.

Collecting the three post-reform observations for each community into vectors $\mathbf{r}_j$ and the corresponding water changes into $\mathbf{x}_j$, we write

$$\mathbf{r}_j = \eta_j \mathbf{x}_j + \boldsymbol{\varepsilon}_j, \quad \eta_j \sim \mathcal{N}(0, \hat{\sigma}_\eta^2). \quad (31)$$

Because each utility change is measured relative to the same 2006 baseline, the errors in $\boldsymbol{\varepsilon}_j$ are correlated across post-reform periods. We model the underlying utility shocks as a stationary AR(1) process and estimate its covariance matrix $\hat{\mathbf{B}}$ by maximum likelihood.¹³

Given these variance components, the posterior mean of the location-specific effect is

$$\hat{\eta}_j^{EB} = \left[\frac{1}{\hat{\sigma}_\eta^2} + \mathbf{x}_j' \hat{\mathbf{B}}^{-1} \mathbf{x}_j \right]^{-1} \mathbf{x}_j' \hat{\mathbf{B}}^{-1} \mathbf{r}_j. \quad (32)$$

The estimator shrinks noisy location-specific responses toward the population mean, with less shrinkage for communities experiencing larger and more informative water changes. Our estimate of each community’s total water value is therefore

$$\hat{\omega}_j = \hat{\omega} + \hat{\eta}_j^{EB}. \quad (33)$$

The resulting values are expressed in utility units. To express them in dollars, we use the estimated coefficient on log housing costs as the numeraire. This gives each community’s willingness to pay per ML of irrigation water, which can be compared directly with the productive water values estimated below.¹⁴

6.2 Productive Value of Water

We construct agricultural producers’ marginal values of water by adapting the production-function estimates of [Rafey \(2023\)](#) to our regional data. Rafey estimates crop-specific production technologies using detailed MDB farm microdata. We take the estimated technological parameters as given rather than re-estimating the agricultural production function.

Following Rafey, crop c production in region j is

$$Y_{jct} = \exp\{\omega_{jct}\} \left[\alpha_c (W_{jct} + \theta_c R_{jct})^{\frac{\sigma_c - 1}{\sigma_c}} + (1 - \alpha_c) K_{jct}^{\frac{\sigma_c - 1}{\sigma_c}} \right]^{\frac{\sigma_c}{\sigma_c - 1} \beta_{cW}} L_{jct}^{\beta_{jctL}}. \quad (34)$$

¹³Appendix Section D provides details of the covariance structure and estimation.

¹⁴Appendix Section F discusses details of this unit conversion.

We take $(\hat{\alpha}_c, \hat{\theta}_c, \hat{\sigma}_c, \hat{\beta}_{cW}, \hat{\beta}_{cL})$ from Rafey’s preferred estimates. Appendix Table 10 reports the crop-specific values used in our calculations.

Two adjustments are required to apply these farm-level estimates to our regional data. First, we allow the effective labor elasticity to differ across SA2s because aggregation across heterogeneous farms can cause regional labor responses to differ from within-farm responses. We parameterize

$$\beta_{jcL} = \hat{\beta}_{cL} \exp\{\delta_j\}, \quad (35)$$

and recover the persistent regional adjustment δ_j from regional labor expenditures and agricultural output, shrinking noisy regional estimates toward the farm-level estimates. Second, conditional on these labor elasticities, we invert Equation 34 to recover the crop-region-year productivity shifter ω_{jct} . Appendix Section E describes both steps in detail.

We then evaluate the marginal product of irrigation water at the observed crop-region-year allocation:

$$\hat{V}_{jct}^P = F_W \left(W_{jct}, R_{jct}, K_{jct}, L_{jct}; \hat{\omega}_{jct}, \hat{\Theta}_c \right), \quad (36)$$

where $\hat{\Theta}_c$ collects the crop-specific production parameters. By the envelope result in Equation 14, this marginal product is the producer’s marginal value of water.

For the market analysis, we require a single producer value for each region rather than separate values for each crop. We aggregate crop-specific marginal values using their observed shares of irrigation water,

$$\hat{V}_{jt}^P = \sum_c \frac{W_{jct}}{\sum_{c'} W_{jc't}} \hat{V}_{jct}^P. \quad (37)$$

This aggregation values a marginal proportional increase in irrigation holding fixed the observed allocation of water across crops.¹⁵

7 Results

7.1 Residential demand parameter estimates and community value of water.

We begin by reporting the residential preference estimates that underlie our measure of community water values. We estimate our demand specification with a panel of SA1-level migration data from 2006-2021 in five year intervals. We set the discount parameter $\beta = 0.95$ and the continuation correction smoothing parameter using Jeffrey’s prior.¹⁶

First-stage estimates: Indirect flow utilities and moving cost. The first-stage estimation recovers indirect flow utility parameters, $\delta_{jt\tau}$, origin SA2 fixed-effects, γ_{jt-1} , and the moving cost per kilometer, κ . The second-stage will decompose indirect utility. Our estimate of moving cost implies that each kilometer imposes a utility cost equivalent to a 2.1 percent increase in rent. At

¹⁵Allowing the crop allocation itself to respond to additional water would require modeling crop choice and within-region water reallocation. Our benchmark instead holds the observed crop mix fixed.

¹⁶See Appendix Section B.4 for more details.

the mean rent in our sample and using the estimated price elasticity in Table 1, this corresponds to approximately AUD \$247 per year for each additional kilometer moved.

Second-stage estimates: Residential preferences. In the second stage, we decompose the recovered flow utilities into observed characteristics and residual local utility. Table 1 reports the resulting preference estimates. Households prefer communities with higher incomes and greater local irrigation, while higher housing costs reduce residential utility. A one-log-point increase in household income raises flow utility by 4.57, while a one-log-point increase in rent reduces flow utility by 1.98.¹⁷ The estimated coefficient on local irrigation is 0.159 and is precisely estimated. Households also prefer communities with a larger stock of long-term residents: the estimated coefficient on tenure stock is 1.43, although it is less precisely estimated.

The instruments strongly predict the endogenous location characteristics. The first-stage F -statistics are 382 for income, 524 for rent, 16 for tenure stock, and 1,137 for irrigation. The estimates imply economically meaningful preferences over the observed attributes of communities while leaving substantial variation in local utility unexplained by these characteristics. We use this remaining variation, together with reform-induced changes in irrigation, to recover heterogeneity in the value communities place on water.

7.2 Community Value of Water

We next estimate the value communities place on retaining local irrigation. Our measure uses households' revealed residential preferences to measure the total effect of irrigation on the attributes of a community that residents value without estimating each channel separately.

We estimate that a ten-percent increase in local irrigation raises non-housing flow utility by 0.567 utility units. This estimate, $\widehat{\omega}$, captures the average effect of irrigation on community utility, including both its conditional value and its effects on other valued attributes of place. The magnitude is substantial at over a quarter of the housing cost preference parameter from Table 1.

Community water values also vary substantially across locations. Figure 4 provides descriptive evidence of this heterogeneity by plotting the dispersion of residual utility changes, r_{jt} , from Equation 29 against instrument-induced water reallocation. Residual utility changes are relatively concentrated among communities experiencing little reallocation and become more dispersed as the magnitude of water changes increases. This pattern is consistent with persistent differences across communities in the utility effect of irrigation.

Our cross-period covariance estimator confirms this heterogeneity. We estimate the variance of persistent community-specific water effects specified in Equation 30 and find that $\widehat{\sigma}_\eta^2 = 0.145$ (s.e. 0.052), corresponding to a standard deviation of $\widehat{\sigma}_\eta = 0.381$.¹⁸ These results indicate substantial variation across communities in the value of local irrigation.

¹⁷Translating our estimates into an own-price elasticity at the SA2 level, which is more comparable to US Census tract sizes, we get a median 10-year elasticity of 2.54 which aligns with estimates from CITE CITE CITE.

¹⁸Appendix Table 9 reports the estimates.

Table 1: Residential Choice Parameters

Dependent Variable:	$\hat{\delta}_{jt\tau}$
Water	0.1594 (0.0400)
log(Income)	4.572 (1.348)
log(Rent)	-1.976 (0.4977)
Tenure stock	1.425 (0.809)
<i>Controls</i>	
Death Rate	Yes
Tenure Status	Yes
SA1	Yes
Water District $\times$ Year	Yes
<i>Fit statistics</i>	
Observations	59,284
R ²	0.2512
F-test (1st stage), Water	1,137.1
F-test (1st stage), log(Income)	382.12
F-test (1st stage), log(Rent)	523.84
F-test (1st stage), Tenure stock	16.35

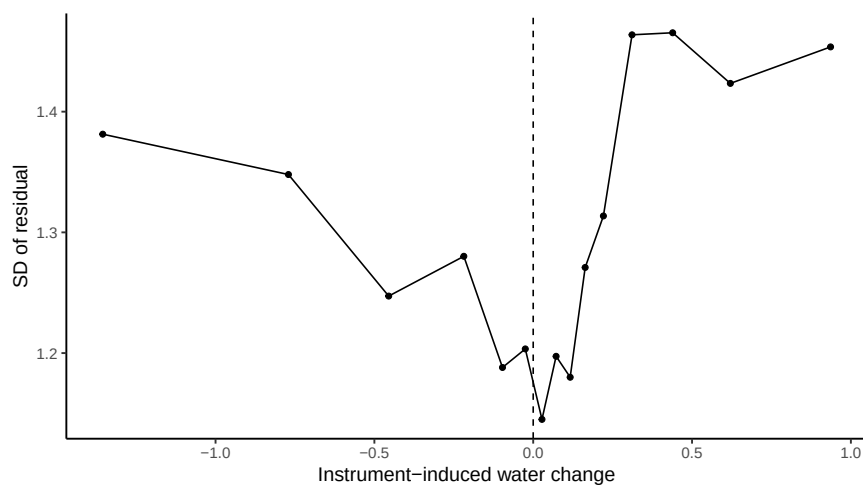
Notes: The dependent variable is flow indirect utility recovered from the first-stage residential choice model. The specification includes tenure-state, destination, and water-region-by-year fixed effects and controls for the local death rate. Income, rent, tenure stock, and irrigation are instrumented as described in Section 6. Standard errors are clustered by destination. First-stage F -statistics for the excluded instruments are reported at the bottom of the table.

We use the Empirical Bayes procedure described in Section 6.1.3 to combine this population heterogeneity with each community’s observed utility responses and recover location-specific water values. Figure 5 plots the resulting distribution after converting utility effects into willingness to pay using households’ estimated marginal utility of housing expenditure.¹⁹

The median community is willing to pay \$117 per ML to retain local irrigation. The interquartile range is \$70–\$183/ML, and the 10th and 90th percentiles are \$43 and \$252/ML. We find that median willingness to pay is around half the average market price of \$235/ML over this period with 13% of communities exhibiting values above this market price.

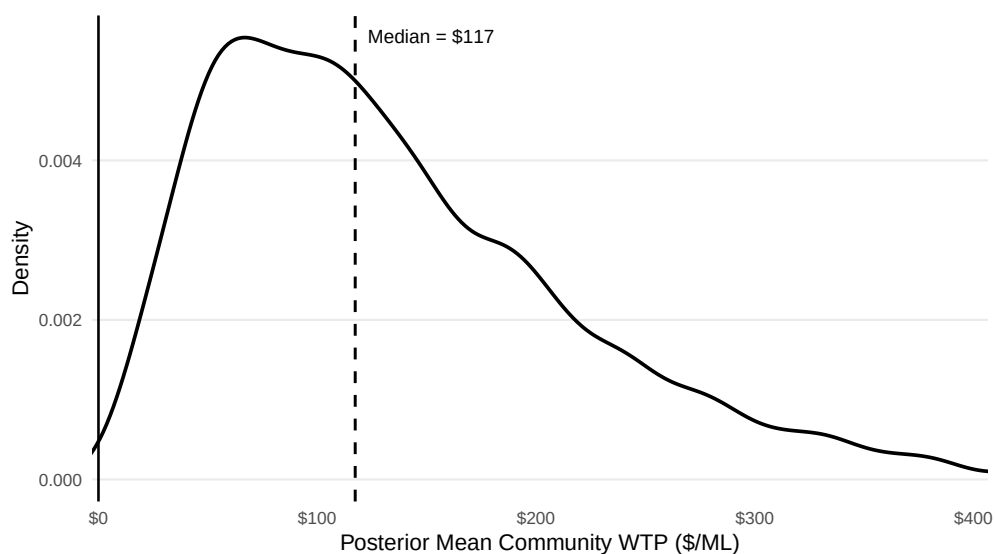
¹⁹Appendix Section F describes the unit conversion.

Figure 4: Heterogeneity in the Effect of Water Reallocation on Local Utility



Notes: The figure plots the standard deviation of residual changes in local utility against instrument-induced changes in irrigation. Residual utility changes remove the average effect of water reallocation. Points report standard deviations within bins of predicted water changes.

Figure 5: Distribution of Community Willingness to Pay



Notes: The figure plots the distribution of posterior mean community willingness to pay for local irrigation across SAIs. Values are expressed in 2021 AUD per ML.

7.3 Agricultural production and the private value of water

We estimate our agricultural production model with a panel of SA2-level data from 2005-2020 in five year intervals. As described in Section 6, we apply the crop-specific production technology estimates from Rafey (2023) to our regional production data. Because these parameters are estimated at the farm level, we allow the labor elasticity to adjust across SA2s when applying the production function to regional aggregates. The mean estimated adjustment, $\exp\{\hat{\delta}_j\}$, is 0.73, with an interquartile range of 0.36–1.27. Appendix Table 12 reports the distribution of these adjustments and the resulting productivity estimates.

We use the estimated production functions to calculate the marginal product of irrigation water for each crop, SA2, and year. We then aggregate across crops within an SA2-year using observed shares of applied irrigation water. This produces the location-year productive value V_{jt}^P used in the market-design counterfactuals. The resulting distribution is highly dispersed and right-skewed. Across SA2-years, the median productive value of water is \$265/ML, with an interquartile range of \$137–\$840/ML. The mean is substantially higher, at \$934/ML, reflecting a long right tail of locations in which the marginal product of water is particularly high.

Much of this dispersion reflects differences across agricultural uses. Median marginal values are \$156/ML for dairy and pasture, \$180/ML for annual crops, and \$1,148/ML for perennial crops. Appendix Table 13 and Appendix Figure 10 report the full crop-specific distributions. Appendix Figure 11 shows that productive values also vary over time with hydrological conditions, rising in dry years and falling in wetter years. For the counterfactual analysis below, we retain the location and year specific productive values rather than collapsing this variation to a single value for each region.

7.4 Community vs. private value for water

The community and productive values we recover are both economically large, but capture distinct consequences of water reallocation. The median community willingness to pay for local irrigation is \$117/ML, compared with a median productive value of \$265/ML across SA2-years. The distributions have substantial overlapping support: the interquartile range of community values is \$70–\$183/ML, while that of productive values is \$137–\$840/ML. In Appendix Figure 12, we depict the distribution of productive gains and community values. Community values can therefore be substantial relative to the private values that govern the allocation of water, even though the productive-value distribution has a considerably longer right tail.

The two values are also largely unrelated across space. After removing year means, the correlation between community and productive values is 0.03. Thus, locations where water generates particularly high agricultural returns are not systematically the same locations where residents place particularly high value on retaining water locally.²⁰

²⁰ Aggregating community values to the SA2-year level, the same geography at which we measure productive values, yields a correlation of 0.08.

We next ask how these two values relate to the direction of water reallocation following market integration. Table 2 relates post-reform changes in local irrigation to productive and community values. Columns 1–3 use the log ratio of post-reform to pre-reform irrigation, while Columns 4–6 consider whether a location experienced a net increase in irrigation.

Table 2: Water Reallocation and the Value of Water

Outcome Model:	Log water ratio			Water gain		
	(1)	(2)	(3)	(4)	(5)	(6)
	OLS	OLS	OLS	Logit	Logit	Logit
Log farmer value	0.122 (0.029)		0.127 (0.029)	0.209 (0.065)		0.222 (0.065)
Log community value		-0.055 (0.048)	-0.082 (0.048)		-0.197 (0.087)	-0.238 (0.089)

Notes: Columns 1-3 regress the log ratio of post to pre-period water availability on estimated productive and community values of water. Columns 4-6 use an indicator for an increase in local water availability as the dependent variable. All specifications include year fixed effects. Standard errors are clustered at the SA2 level.

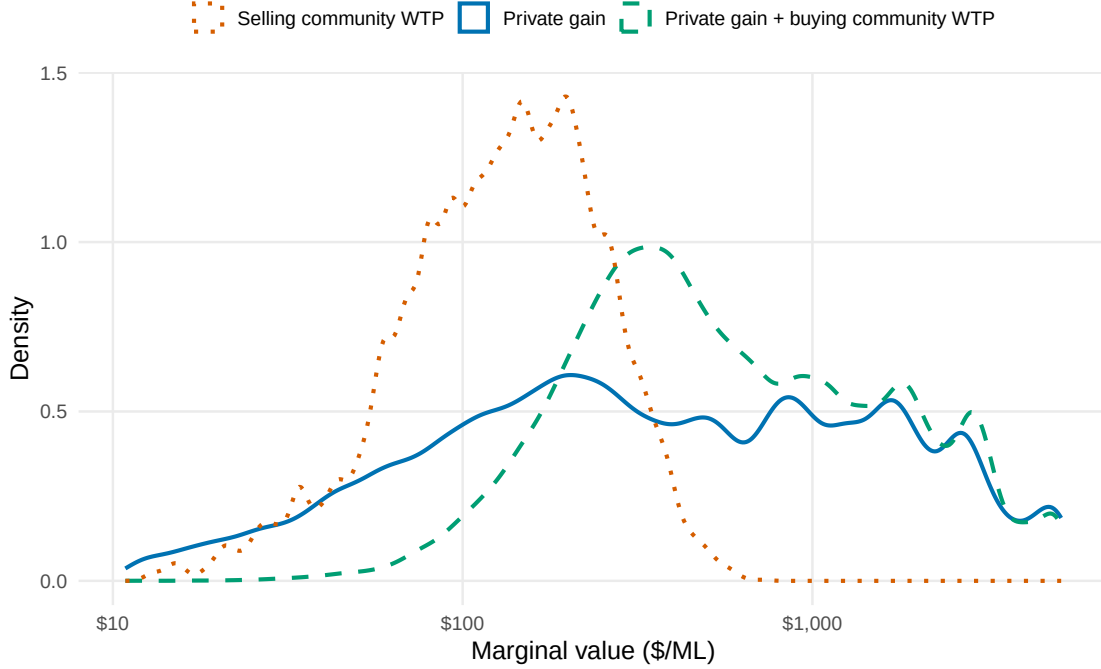
Water reallocation is strongly associated with productive values. In the joint specification, a 10% higher productive value is associated with approximately 1.3% more irrigation relative to the pre-reform period. In contrast, a 10% higher community value is associated with approximately 0.8% less irrigation, although this relationship is less precisely estimated. The logit specifications exhibit the same pattern: productive values are positively associated with the probability that a location gained water, while community values are negatively associated with that probability.

These results show that productive and community values represent distinct dimensions of the value of local water. Market integration moved water toward locations where its productive value was higher, but productive values contain little information about where the community value of water is high. Whether the private gains generated by a transfer are sufficient to compensate the communities it affects therefore depends on the joint distribution of buyer and seller productive and community values. We turn to this question in the next section.

8 Counterfactuals: Local Control and Market Reallocation

We now use the estimated community and productive values to quantify the market-design tradeoffs introduced in Section 2. Consider a potential transfer of one ML of water from a seller s to a buyer b . The private gain from the transfer is $G_{sb}^P = V_b^P - V_s^P$. In the market benchmark, the buyer and seller can trade whenever $G_{sb}^P > 0$, without obtaining the consent of either community. We normalize the set of these privately gainful trade opportunities to one and ask how many continue to occur when selling communities are given control over whether water can leave.

Figure 6: Comparing trade surplus to community values for water



Notes: The figure compares the distributions of community and private values associated with potential water transfers. Selling community WTP is the selling community's willingness to pay to retain one ML of water locally. Private gain is the difference between the buying and selling farmers' productive values of one ML of water, $G_{sb}^P = V_b^P - V_s^P$. Private gain + buying community WTP adds the buying community's willingness to pay for the local benefits of receiving the water, V_b^C , to the private gain from trade. The horizontal axis is shown on a logarithmic scale and values are expressed in AUD per ML.

As in Section 2, the consequences of community veto rights depend on two features of the environment. The first is the surplus available to secure community consent. The buyer and seller can use the private gain G_{sb}^P ; if the benefit to the buying community can also be mobilized, available surplus rises to $G_{sb}^P + V_b^C$. The second is information about the selling community's value. When V_s^C is known, compensation can be tailored to the amount required to secure consent. When it is private information, traders must make an offer without knowing the community's reservation value.

Figure 6 illustrates the magnitudes underlying these comparisons. The dotted distribution reports the compensation required to secure selling-community consent, V_s^C , while the solid distribution reports the private gains available from trade, G_{sb}^P . There is substantial overlap: many privately gainful trades generate enough surplus to compensate the selling community, while others do not. The dashed distribution adds the benefit to the buying community, V_b^C , shifting available surplus to the right. All three distributions are highly dispersed, making both their joint realization and the information available about community values consequential for which trades proceed.

The marginal distributions alone do not determine market performance. Whether a particular trade can secure community consent depends on the joint realization of V_s^P , V_b^P , V_s^C , and V_b^C . We construct this joint distribution by simulating potential bilateral trades, randomly drawing buyer–

seller pairs from locations in our data without replacement. This allows us to evaluate alternative market rules without imposing a model of how buyers and sellers are matched.²¹

Community Veto under Full Information We first grant selling communities veto rights and suppose that all values are observed. If compensation can be financed only from the private gains generated by the trade, a transfer proceeds whenever

$$G_{sb}^P \geq V_s^C. \quad (38)$$

Full information therefore isolates a surplus constraint: trades that fail this condition do not generate sufficient private gains to make the selling community whole.

Table 3 reports the results. With private surplus alone, 69.0% of privately gainful trade opportunities continue to occur when selling communities receive veto rights. Selling communities receive average compensation of \$159/ML among implemented trades. The remaining 31.0% cannot generate sufficient private surplus to compensate the selling community; these unrealized trades have average private gains of \$86/ML.

The result changes substantially when benefits to the buying community can also be mobilized. A trade can then secure selling-community consent whenever

$$G_{sb}^P + V_b^C \geq V_s^C. \quad (39)$$

Under this condition, 90.2% of privately gainful trade opportunities are implemented. Thus, much of the trade that would be blocked when compensation is limited to the transacting parties can be preserved by drawing on the benefits that reallocation creates for the receiving community.

Table 3: Summary of bilateral trading protocols

Market design	Buying surplus	Trades (%) implemented	Avg. Payment (\$/ML)	Avg. lost gain (\$/ML)
No compensation	—	100.0	0	—
Full information	No	69.0	159	86
Full information	Yes	90.2	163	66
Trader TIOLI	No	50.8	267	273
Trader TIOLI	Yes	68.1	259	332

Notes: The table reports outcomes under alternative bilateral trading protocols. Trade opportunities are transfers with positive private gains. “Buying surplus” indicates whether the buying community’s value can also finance compensation. “Trades implemented” is the share of trade opportunities that proceed. “Avg. payment” is mean compensation to selling communities among implemented trades, and “Avg. lost gain” is the mean private gain among unrealized trades. Under full information, selling-community values are observed and compensation is tailored to secure consent. Under Trader TIOLI, these values are privately known and traders make an optimal take-it-or-leave-it offer. Monetary values are in AUD per ML.

²¹Observed matching in the MDB reflects heterogeneous trading rules, institutional constraints, and transaction costs that are difficult to model without imposing substantial additional structure (?).

Community Veto under Private Information Full information abstracts from a second friction emphasized in Section 2: the compensation required to secure a community’s consent may itself be private information. We therefore suppose that the selling community observes V_s^C , while traders know only its population distribution. For each potential trade, the buyer and seller make a take-it-or-leave-it compensation offer. A higher offer increases the probability that the community consents but reduces the surplus retained by the traders when the offer is accepted. We allow traders to choose the offer that maximizes their expected surplus.²²

Private information substantially reduces the amount of trade that survives community veto. When compensation is financed only from private gains, the share of trade opportunities implemented falls from 69.0% under full information to 50.8% under the Trader TIOLI mechanism. Average compensation among implemented trades rises from \$159/ML to \$267/ML. Communities whose values lie below the optimal offer receive information rents, while some communities whose values could have been compensated from the available surplus reject offers below their reservation values.

The foregone trades illustrate the cost of this information friction. Trades that fail under full information with private surplus have average private gains of only \$86/ML. Under private information, unrealized trades have average private gains of \$273/ML. Thus, private information prevents some trades with substantial productive gains even though sufficient surplus could, under full information, have been used to secure community consent.

Uniform Compensation Community veto is not the only institution available to address the local effects of trade. We finally consider a uniform compensation requirement that does not give selling communities the right to block a transfer. Every implemented trade instead makes the same payment c to the selling community. A trade proceeds whenever the surplus available from the transaction is sufficient to make this payment, regardless of whether c exceeds the community’s value V_s^C . Uniform compensation therefore does not require eliciting each community’s value as a condition for trade, but it also does not guarantee that affected communities are made whole.

Figure 7 traces the resulting tradeoff as the required payment varies. The horizontal axis reports the share of privately gainful trade opportunities that continue to be implemented, while the vertical axis reports the share of selling communities associated with implemented trades that receive less than their estimated value. At a zero payment, all privately gainful trades proceed but essentially all selling communities are undercompensated. Higher required payments reduce undercompensation but also eliminate trades whose available surplus is insufficient to finance the payment. The solid curve restricts compensation to the private gains from trade, while the dashed curve additionally allows buying-community surplus to finance the required payment.

The community-veto outcomes provide useful benchmarks for this tradeoff. With private surplus alone, full-information community veto implements 69.0% of trade opportunities while ensuring that every selling community associated with an implemented trade is fully compensated. A uniform

²²Appendix Section G provides the details of the mechanism. The optimal offer is the procurement analogue of a monopolist’s optimal posted price (Myerson, 1981, Bulow & Roberts, 1989).

Figure 7: Uniform Compensation and Market Reallocation



Notes: The figure traces outcomes under a uniform compensation requirement as the required payment to selling communities varies. The horizontal axis reports the share of privately gainful trade opportunities implemented, and the vertical axis reports the share of selling communities associated with implemented trades whose estimated willingness to pay exceeds the required payment. The solid curve allows compensation to be financed from private gains alone; the dashed curve additionally allows buying-community surplus to be mobilized. Circles denote the full-information community-veto benchmarks and triangles denote the private-information TIOLI benchmarks. Dollar labels report the uniform payment that produces the corresponding trade rate.

payment of \$152/ML produces approximately the same amount of trade, but leaves about half of selling communities undercompensated. Similarly, the private-information TIOLI mechanism implements 50.8% of trade opportunities with no undercompensated communities. A uniform payment must rise to \$333/ML to reduce trade to approximately the same level, yet some selling communities remain undercompensated. The same pattern holds when benefits to the buying community can be mobilized, but we see a rightward shift in the curve given that additional surplus allows for more feasible trades.

These counterfactuals show that the effect of community control on market reallocation depends on more than the magnitude of local losses. Under full information, most privately gainful trades can still proceed while fully compensating selling communities, particularly when the benefits accruing to buying communities can also be mobilized. Private information creates an additional friction: compensation must leave information rents to some communities, while other trades fail even when their gains would have been sufficient to secure consent under full information. Uniform compensation avoids conditioning trade on community consent, but cannot target payments to heterogeneous community values and therefore preserves trade only by tolerating some undercompensation. These results highlight the central market-design tradeoff created by local control: the extent to which markets can accommodate the interests of affected nonparticipants depends jointly on the surplus

generated by reallocation, who can contribute that surplus, and the information available to direct compensation where it is most valuable.

9 Conclusion

Market integration creates gains by reallocating resources toward higher-value uses, but the same reallocation can impose substantial consequences on people outside the transaction. When affected communities have the power to block trade, these local consequences become part of the market-design problem. We study this tension in Australia’s Murray–Darling Basin, where water-market integration generated large productive gains while moving irrigation activity across rural communities.

We find that communities place substantial and highly heterogeneous value on retaining irrigation. These values are only weakly related to the productive value of water, so the market reveals where gains from reallocation can be created but provides little information about where compensation is most valuable. This distinction matters when communities receive control over trade. Under full information, many profitable trades can still proceed while fully compensating affected communities, particularly when benefits accruing to buying communities can also be mobilized. Private information makes local control more restrictive because compensation cannot be perfectly targeted. Uniform compensation avoids conditioning trade on community consent, but cannot target heterogeneous community values and therefore leaves some communities undercompensated.

The broader lesson is that local control need not simply be understood as a barrier to efficient markets. Giving affected nonparticipants rights over transactions brings values that markets otherwise ignore into the bargain. How costly those rights are for market reallocation depends on whether the gains from trade are large enough to accommodate those values, whether gains accruing outside the transaction can be mobilized, and whether compensation can be targeted when local values are privately known. These forces arise whenever market integration reallocates economic activity across place or use. Understanding them can help distinguish settings in which local control primarily blocks gains from trade from those in which bargaining can preserve reallocation while giving affected communities a meaningful role in whether it occurs.

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A Measuring local water use for irrigation

Our empirical analysis requires a measure of irrigation water availability at the SA1 level. Administrative records on water allocations and trades are not available at this spatial resolution. We therefore construct a measure using satellite estimates of actual evapotranspiration (ET). This appendix describes the construction of the measure and validates it against administrative measures of water allocations and applied irrigation water.

Residual evapotranspiration. Actual ET reflects both irrigation and naturally available water. To avoid mechanically attributing evapotranspiration from rivers and other surface waterways to irrigation, our preferred measure excludes pixels within 500 meters of major waterways. We also construct the measure using exclusion bands of 0, 100, and 250 meters and find that our results are robust to these alternative definitions. Actual ET also responds to rainfall and atmospheric conditions. We isolate variation in ET unexplained by these meteorological conditions.

For each SA1 j , calendar month m , and year t , we estimate

$$ET_{jmt} = \rho_0 Rain_{jmt} + \rho_1 Rain_{jm,t-1} + \psi PET_{jmt} + \gamma_{jm} + \varepsilon_{jmt}, \quad (40)$$

where ET_{jmt} is actual evapotranspiration per hectare, $Rain_{jmt}$ is contemporaneous rainfall, $Rain_{jm,t-1}$ is lagged rainfall, and PET_{jmt} is potential evapotranspiration. The SA1-by-calendar-month fixed effects γ_{jm} absorb persistent local differences in climate, vegetation, and seasonality. We define residual ET as

$$\widehat{\varepsilon}_{jmt} = ET_{jmt} - \widehat{ET}_{jmt}. \quad (41)$$

Positive residual ET therefore indicates that observed evapotranspiration is greater than predicted by local weather conditions and persistent seasonal differences.

Annual water availability. We construct our preferred annual measure using residual ET during the dry months, October through March. Irrigation should contribute more strongly to ET during these months because naturally available water is relatively scarce. For SA1 j in year t , we define

$$W_{jt} = \sum_{m \in \mathcal{M}^{Dry}} \widehat{\varepsilon}_{jmt}, \quad \mathcal{M}^{Dry} = \{\text{October}, \dots, \text{March}\}. \quad (42)$$

Thus, W_{jt} measures annual evapotranspiration in excess of that predicted by weather during the period in which irrigation is most likely to be responsible for this excess water use. Because ET is measured per unit of land, W_{jt} is an intensive-margin measure of local irrigation activity.

For the event-study analysis, we normalize water so that one unit corresponds to 0.096 ML/ha of residual ET. This is approximately a 10 percent change in irrigation intensity at the average SA1. The event-study coefficients can therefore be interpreted as the effect of approximately a 10 percent change in local irrigation activity.

Table 4: Administrative Water Allocations and Residual Evapotranspiration

Table 5: Regional Residual ET versus Allocation Volume

Residual ET Model:	Annual (1)	Dry (2)	Wet (3)
<i>Variables</i>			
Total allocation	0.7312* (0.3757)	0.6511* (0.3286)	-0.0279 (0.1372)
<i>Fixed-effects</i>			
Water District	Yes	Yes	Yes
Year	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	440	440	440
R ²	0.31945	0.78145	0.93048

Notes: The dependent variables are annual, dry-season (October–March), and wet-season residual evapotranspiration, respectively. The explanatory variable is total administrative water allocation. All specifications include water-region and year fixed effects. Standard errors are clustered at the water-region level.

Water exposure. For specifications that require a single measure of the reallocation induced by the water-market reform, we compare average water availability before and after 2007. Specifically, we define

$$\Delta W_{j,t} = \bar{W}_{j,\text{post}\leq t} - \bar{W}_{j,\text{pre}}, \quad (43)$$

where $\bar{W}_{j,\text{pre}}$ and $\bar{W}_{j,\text{post}\leq t}$ denote average dry-season residual ET before and after the 2007 reform up to year t , respectively. Positive values of $\Delta W_{j,t}$ correspond to locations in which irrigation activity increased relative to the pre-reform period, while negative values correspond to locations in which it declined.

Validation with administrative water use. We validate our satellite-based measure using administrative records of annual water allocations at the water-region level. Because these data directly measure the amount of water made available to users, but are observed at a substantially coarser geography than our SA1 analysis, they provide a useful external validation of the residual ET measure.

We aggregate residual ET to the corresponding water region and estimate

$$W_{rt}^{ET} = \beta A_{rt} + \alpha_r + \lambda_t + u_{rt}, \quad (44)$$

where W_{rt}^{ET} is residual ET in water region r , A_{rt} is total administrative water allocation, and α_r and λ_t are region and year fixed effects. Standard errors are clustered by water region.

Table 4 compares residual ET constructed over the full year, the dry season, and the wet season.

Administrative allocations are positively related to both annual and dry-season residual ET, with a particularly strong relationship during the dry season. A one-unit increase in total allocation is associated with a 0.65-unit increase in dry-season residual ET. In contrast, allocations have essentially no relationship with wet-season residual ET: the estimated coefficient is -0.03 . This seasonal contrast supports our interpretation that residual ET during the dry season primarily captures irrigation water rather than unobserved natural variation in evapotranspiration, and motivates our preferred October–March measure.

We further decompose total water availability into direct allocations, environmental buybacks, and net agricultural transfers. Appendix Table 14 reports the results. The positive relationship with dry-season residual ET is driven primarily by direct water allocations. Environmental buybacks and net agricultural transfers are considerably noisier and are estimated imprecisely at this level of aggregation. As in the aggregate specification, none of these components meaningfully predicts wet-season residual ET.

Finally, we conduct a similar validation using ABS measures of applied irrigation water at the finer SA2 level. We view these data as a noisier measure of annual local water availability, but they allow validation closer to the spatial scale of our empirical analysis. The results are similar: dry-season residual ET is positively associated with observed applied irrigation water, whereas wet-season residual ET is not. Appendix Table 16 reports these results.

B Derivation of the Dynamic Residential Choice Estimator

This appendix derives the first-stage residential choice estimator described in Section 6. We first use the Hotz–Miller inversion to express current relative choice probabilities in terms of flow utility, moving costs, next-period conditional choice probabilities, and continuation values. We then exploit the renewal structure of the outside option to eliminate the remaining continuation values and obtain the estimating equation used in the first-stage likelihood.

B.1 Dynamic choice probabilities

A household enters period t with individual state

$$x_{t-1} = (j_{t-1}, \tau_{t-1}), \quad (45)$$

where j_{t-1} denotes the household’s previous location and τ_{t-1} its tenure in that location. The household chooses destination j and receives flow utility

$$u_{ht}(j \mid x_{t-1}) = \bar{u}_{jt\tau} + MC_t(j, j_{t-1}) + \varepsilon_{hjt}, \quad (46)$$

where $\bar{u}_{jt\tau}$ is mean flow utility from choosing destination j in period t and entering tenure state τ , $MC_t(j, j_{t-1})$ captures the utility cost of moving from the household’s previous location, and ε_{hjt} is an idiosyncratic Type-I extreme value preference shock. We suppress aggregate states throughout

the derivation for notational convenience.

Under the conditional-independence assumptions described in the main text, define the expected value of entering period t in state x_{t-1} , prior to the realization of the idiosyncratic shocks, as

$$\bar{V}_t(x_{t-1}) = \log \left[\sum_{j \in \mathcal{J}} \exp \{v_t(j, x_{t-1})\} \right], \quad (47)$$

where the choice-specific value is

$$v_t(j, x_{t-1}) = \bar{u}_{jt\tau} + MC_t(j, j_{t-1}) + \beta \bar{V}_{t+1}(x'_j), \quad (48)$$

and

$$x'_j \equiv (j, \tau_t(j, \tau_{t-1})) \quad (49)$$

is the household state induced by choosing j .

The corresponding conditional choice probability is

$$\mathbb{P}_t(j \mid x_{t-1}) = \frac{\exp\{v_t(j, x_{t-1})\}}{\sum_{\ell \in \mathcal{J}} \exp\{v_t(\ell, x_{t-1})\}}. \quad (50)$$

We normalize the flow utility of the composite outside option j_0 to zero,

$$\bar{u}_{j_0 t} = 0. \quad (51)$$

Taking the log odds of choosing an inside location j relative to the outside option gives

$$\begin{aligned} \log \frac{\mathbb{P}_t(j \mid x_{t-1})}{\mathbb{P}_t(j_0 \mid x_{t-1})} &= \bar{u}_{jt\tau} + MC_t(j, j_{t-1}) - MC_t(j_0, j_{t-1}) \\ &\quad + \beta [\bar{V}_{t+1}(x'_j) - \bar{V}_{t+1}(x'_{j_0})]. \end{aligned} \quad (52)$$

The remaining difference in continuation values prevents direct estimation.

B.2 Hotz–Miller inversion and renewal

For any state x and reference action a , the Hotz–Miller inversion implies

$$\bar{V}_{t+1}(x) = v_{t+1}(a, x) - \log \mathbb{P}_{t+1}(a \mid x). \quad (53)$$

We apply this identity to each continuation value in Equation (52), using the outside option j_0 as the reference action:

$$\begin{aligned} \bar{V}_{t+1}(x'_j) - \bar{V}_{t+1}(x'_{j_0}) &= v_{t+1}(j_0, x'_j) - v_{t+1}(j_0, x'_{j_0}) \\ &\quad - \log \frac{\mathbb{P}_{t+1}(j_0 \mid x'_j)}{\mathbb{P}_{t+1}(j_0 \mid x'_{j_0})}. \end{aligned} \quad (54)$$

Because the outside option has normalized flow utility, the difference in choice-specific values is

$$v_{t+1}(j_0, x'_j) - v_{t+1}(j_0, x'_{j_0}) = MC_{t+1}(j_0, j) - MC_{t+1}(j_0, j_0) + \beta [\bar{V}_{t+2}(x''_{j,j_0}) - \bar{V}_{t+2}(x''_{j_0,j_0})]. \quad (55)$$

The outside option provides a renewal action that eliminates the final continuation-value difference. Tenure is a payoff-relevant individual state only within the modeled MDB locations. Choosing the composite outside option resets tenure:

$$\tau_t(j_0, \tau) = 0 \quad \forall \tau. \quad (56)$$

Thus, regardless of whether the household chooses j or j_0 in period t , choosing j_0 in period $t + 1$ produces the same state in period $t + 2$:

$$x''_{j,j_0} = x''_{j_0,j_0} = (j_0, 0). \quad (57)$$

It follows that

$$\bar{V}_{t+2}(x''_{j,j_0}) - \bar{V}_{t+2}(x''_{j_0,j_0}) = 0. \quad (58)$$

The outside option therefore generates finite dependence: after one additional outside choice, the states induced by the two current choices coincide. This eliminates continuation values from period $t + 2$ onward without requiring us to solve the household value function.

Substituting into Equation (52) gives

$$\begin{aligned} \log \frac{\mathbb{P}_t(j \mid x_{t-1})}{\mathbb{P}_t(j_0 \mid x_{t-1})} &= \bar{u}_{jt\tau} + MC_t(j, j_{t-1}) - MC_t(j_0, j_{t-1}) \\ &\quad + \beta [MC_{t+1}(j_0, j) - MC_{t+1}(j_0, j_0)] \\ &\quad - \beta \log \frac{\mathbb{P}_{t+1}(j_0 \mid x'_j)}{\mathbb{P}_{t+1}(j_0 \mid x'_{j_0})}. \end{aligned} \quad (59)$$

Equation (59) is the key estimating relationship. The dynamic component of the household problem is summarized by next-period exit probabilities, while all continuation values have been eliminated.

B.3 Moving costs and first-stage flow utility

We next map Equation (59) into the parameterization used in estimation. Normalize the utility cost of remaining in the outside option,

$$MC_{t+1}(j_0, j_0) = 0. \quad (60)$$

The remaining moving-cost terms are

$$\underbrace{MC_t(j, j_{t-1})}_{\text{origin-destination}} - \underbrace{MC_t(j_0, j_{t-1})}_{\text{origin}} + \underbrace{\beta MC_{t+1}(j_0, j)}_{\text{destination}}. \quad (61)$$

The first term varies jointly across origins and destinations. Rather than introducing unrestricted origin-by-destination parameters, which would absorb much of the variation used to identify destination preferences, we parameterize this component using distance:

$$MC_t(j, j_{t-1}) = \kappa \text{Dist}(j, j_{t-1}), \quad (62)$$

where the sign of κ follows our utility convention.

The second term depends only on the household's origin and is absorbed by origin fixed effects, $\gamma_{j_{t-1}}$. The third depends only on the destination and is not separately identified from destination flow utility in the first stage. We therefore absorb it into the destination-period-tenure utility $\delta_{jt\tau}$. Because this component is time invariant, it is subsequently absorbed by destination fixed effects in the second-stage decomposition of $\hat{\delta}_{jt\tau}$.

The first-stage choice index can therefore be written as

$$\delta_{jt\tau} + \gamma_{j_{t-1}} + \kappa \text{Dist}(j, j_{t-1}) - \beta \log \frac{\mathbb{P}_{t+1}(j_0 \mid x'_j)}{\mathbb{P}_{t+1}(j_0 \mid x'_{j_0})}. \quad (63)$$

The first-stage likelihood estimates the destination-period-tenure utilities $\delta_{jt\tau}$, origin fixed effects $\gamma_{j_{t-1}}$, and distance parameter κ , taking the estimated next-period exit probabilities as given. The recovered $\hat{\delta}_{jt\tau}$ are the tenure-specific flow utilities that enter the second-stage decomposition described in Section 6.1.

B.4 Regularization of exit conditional choice probabilities

The preceding estimator requires empirical CCPs only for the exit probabilities entering Equation 63. Because these probabilities enter logarithmically, a raw empirical exit probability of zero would make the continuation correction undefined.

Let $s = (k, t, x)$ index a state cell, let N_s denote the number of households making a choice from that state, and let N_s^0 denote the number choosing the outside option. The raw empirical exit probability is

$$\hat{q}_s^0 = \frac{N_s^0}{N_s}. \quad (64)$$

We regularize these probabilities using a Beta-Binomial model,

$$q_s^0 \sim \text{Beta}(\alpha_q, \beta_q), \quad (65)$$

$$N_s^0 \mid N_s, q_s^0 \sim \text{Binomial}(N_s, q_s^0). \quad (66)$$

The resulting posterior is

$$q_s^0 \mid N_s^0, N_s \sim \text{Beta}(N_s^0 + \alpha_q, N_s - N_s^0 + \beta_q), \quad (67)$$

and we use its posterior mean,

$$\hat{q}_s^0 = \frac{N_s^0 + \alpha_q}{N_s + \alpha_q + \beta_q}. \quad (68)$$

Equivalently,

$$\hat{q}_s^0 = \frac{N_s}{N_s + \alpha_q + \beta_q} \hat{q}_s^0 + \frac{\alpha_q + \beta_q}{N_s + \alpha_q + \beta_q} \frac{\alpha_q}{\alpha_q + \beta_q}. \quad (69)$$

The regularized estimate therefore shrinks the raw empirical exit probability toward the prior mean $\alpha_q/(\alpha_q + \beta_q)$, with the amount of shrinkage declining in the number of observations in the state cell. We set $(\alpha_q, \beta_q) = (1/2, 1/2)$ in the baseline specification.

This regularization is required only for the exit CCPs used to construct the continuation correction. The likelihood in Equation 26 is evaluated directly from observed destination counts. A destination with zero observed choices from a particular state therefore contributes zero to the corresponding likelihood term and does not require the construction or smoothing of a positive empirical destination CCP.

C Instruments for Residential Location Characteristics

This section describes the instruments used for household income, housing costs, and the local stock of long-term residents in the second-stage residential-choice estimation.

C.1 Industry shift-share instrument for income

We construct a shift-share measure of predicted local employment growth using each SA1's industry composition in 2001. We first map employment by industry from 2001 Census Districts to SA1s and calculate the share of SA1 j 's employment in industry c ,

$$s_{jc,2001} = \frac{L_{jc,2001}}{\sum_{c'} L_{jc',2001}}.$$

For each subsequent Census year $t \in \{2006, 2011, 2016, 2021\}$, we calculate the growth of national employment in the corresponding industry relative to 2001,

$$g_{ct} = \frac{L_{ct}^{\text{AUS}} - L_{c,2001}^{\text{AUS}}}{L_{c,2001}^{\text{AUS}}}.$$

The instrument used in our baseline specification is

$$Z_{jt}^I = \sum_c s_{jc,2001} g_{ct}.$$

It predicts differential changes in local labor-market conditions from predetermined industry composition and subsequent national industry growth. The national growth rates in this baseline construction include employment in the focal SA1.

C.2 Predicted retention instrument for long-term residence

We instrument the local stock of long-term residents with predicted retention based on the SA1's lagged age-by-tenure composition and retention patterns outside its own SA2. For each SA1 j , age

group a , and Census year t , let $s_{ja,t-5}^{1-4}$ and $s_{ja,t-5}^{5+}$ be the shares of the entire lagged SA1 population in the corresponding age-by-tenure cells. These shares are constructed from the preceding Census and shifted forward five years to align them with the outcome year.

For each age group and year, we calculate the shares of individuals observed in the one-to-four-year and five-or-more-year residence categories using migration histories from all SA2s other than the SA2 containing j . Denote these leave-one-SA2-out shares by $r_{a,t,-d(j)}^{1-4}$ and $r_{a,t,-d(j)}^{5+}$. The instrument is

$$Z_{jt}^T = \sum_a \left[s_{ja,t-5}^{1-4} r_{a,t,-d(j)}^{1-4} + s_{ja,t-5}^{5+} r_{a,t,-d(j)}^{5+} \right].$$

Thus, variation in the instrument comes from differences in lagged local age-by-tenure composition interacted with the residence patterns of otherwise similar age groups outside the focal SA2. Excluding the entire SA2 prevents the focal SA1's own migration outcomes from entering the retention rates applied to its exposure shares.

C.3 Demographic instrument for housing costs

Housing costs may respond to changes in local demand that also affect residential utility. To isolate variation in housing costs driven by predetermined demographics, we construct an instrument from each SA1's lagged age composition and the national age distribution of deaths. Let $s_{ja,t-5}$ be the share of residents in age group a in SA1 j five years before year t , and let q_a be the share of deaths nationally that occur in that age group. The instrument is

$$Z_{jt}^p = \sum_a s_{ja,t-5} q_a. \quad (70)$$

National weights are common across locations, so variation in Z_{jt}^p reflects differences in the age composition of local residents. Locations with greater exposure to age groups associated with deaths may experience different rates of housing turnover and, consequently, different pressure on rents. Lagging the local age shares separates this predicted demographic pressure from contemporaneous changes in housing costs. The identifying restriction is that, conditional on the controls and fixed effects in the residential-preference specification, this demographic exposure affects recovered flow utility through housing costs rather than through other changes in local amenities.

D Empirical Bayes Covariance Structure

This section describes the covariance structure used in the Empirical Bayes estimator of location-specific community water values. After removing the estimated mean response to water, residual utility changes satisfy

$$r_{jt} = \eta_j \Delta W_{jt} + e_{jt}, \quad (71)$$

where η_j is the persistent location-specific water effect and e_{jt} captures transitory changes in utility.

Because each post-reform utility change is measured relative to the same 2006 baseline, the

errors e_{jt} are mechanically correlated across periods. We model this correlation using an underlying utility shock in levels. Let $q \in \{0, 1, 2, 3\}$ index the 2006, 2011, 2016, and 2021 observation periods, respectively. We assume that the level shock ν_{jq} follows a stationary AR(1) process,

$$\text{Cov}(\nu_{jq}, \nu_{jr}) = \sigma_\nu^2 \rho^{|q-r|}, \quad 0 < \rho < 1. \quad (72)$$

The three observed long-difference errors are

$$\boldsymbol{\varepsilon}_j = \begin{pmatrix} \nu_{j1} - \nu_{j0} \\ \nu_{j2} - \nu_{j0} \\ \nu_{j3} - \nu_{j0} \end{pmatrix}. \quad (73)$$

Let $\boldsymbol{\Sigma}_\nu(\rho, \sigma_\nu)$ denote the 4×4 covariance matrix implied by Equation (72), and define

$$\mathbf{A} = \begin{pmatrix} -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}.$$

The covariance matrix of the three long-difference errors is therefore

$$\mathbf{B}(\rho, \sigma_\nu) \equiv \text{Var}(\boldsymbol{\varepsilon}_j) = \mathbf{A} \boldsymbol{\Sigma}_\nu(\rho, \sigma_\nu) \mathbf{A}'. \quad (74)$$

This transformation accounts both for serial correlation in the underlying utility shocks and for the covariance induced by the common 2006 baseline.

To estimate the covariance parameters, collect each community's three residual utility changes in $\mathbf{r}_j$ and its corresponding water changes in $\mathbf{x}_j$. We assume

$$\mathbf{r}_j = \eta_j \mathbf{x}_j + \boldsymbol{\varepsilon}_j, \quad \eta_j \sim \mathcal{N}(0, \hat{\sigma}_\eta^2), \quad (75)$$

where $\hat{\sigma}_\eta^2$ is obtained from the cross-period second-moment estimator described in Section 6.1.3. Integrating over η_j gives

$$\mathbf{r}_j \sim \mathcal{N}(\mathbf{0}, \mathbf{B}(\rho, \sigma_\nu) + \hat{\sigma}_\eta^2 \mathbf{x}_j \mathbf{x}_j'). \quad (76)$$

We estimate ρ and σ_ν by maximizing this marginal likelihood across communities.

Conditional on the estimated covariance matrix $\hat{\mathbf{B}}$, normal conjugacy gives the posterior variance and mean of the location-specific effect:

$$V_j^{EB} = \left[\frac{1}{\hat{\sigma}_\eta^2} + \mathbf{x}_j' \hat{\mathbf{B}}^{-1} \mathbf{x}_j \right]^{-1}, \quad (77)$$

$$\hat{\eta}_j^{EB} = V_j^{EB} \mathbf{x}_j' \hat{\mathbf{B}}^{-1} \mathbf{r}_j. \quad (78)$$

Communities with more informative water changes have larger $\mathbf{x}_j' \hat{\mathbf{B}}^{-1} \mathbf{x}_j$, yielding smaller posterior variance and less shrinkage toward the population mean.

E Construction of Regional Productive Water Values

This appendix describes how we adapt the farm-level production-function estimates of [Rafey \(2023\)](#) to our regional agricultural data. We do not re-estimate the underlying crop-specific production technologies. Instead, we take Rafey’s preferred technological parameters as given and recover the regional objects required to evaluate the marginal product of irrigation water at observed regional allocations.

E.1 Production Technology

For crop c in region j and year t , production is

$$Y_{jct} = \exp\{\omega_{jct}\} \left[\alpha_c (W_{jct} + \theta_c R_{jct})^{\frac{\sigma_c-1}{\sigma_c}} + (1 - \alpha_c) K_{jct}^{\frac{\sigma_c-1}{\sigma_c}} \right]^{\frac{\sigma_c}{\sigma_c-1} \beta_{cW}} L_{jct}^{\beta_{cL}}, \quad (79)$$

where W_{jct} is irrigation water, R_{jct} is rainfall, K_{jct} is cultivated land, L_{jct} is labor, and ω_{jct} is crop-region-year productivity. Since we do not separately observe output prices or expenditures on intermediate inputs, ω_{jct} absorbs variation in these components in addition to physical productivity. We therefore interpret ω_{jct} as a crop-region-year revenue productivity shifter, consistent with our objective of recovering the marginal value of irrigation water.

The crop-specific parameters α_c , θ_c , σ_c , β_{cW} , and β_{cL} are taken from [Rafey \(2023\)](#). The parameter α_c determines the distributional share of water relative to land, θ_c converts rainfall into effective water, σ_c governs substitution between effective water and land, and β_{cW} and β_{cL} govern the output elasticities of the water-land aggregate and labor.

The regional data differ from the farm microdata used to estimate these parameters in one important respect. Aggregating across heterogeneous farms can cause the response of total regional labor to differ from the within-farm labor elasticity. We therefore allow the effective regional labor elasticity to deviate from the farm-level estimate:

$$\beta_{jcL} = \hat{\beta}_{cL} \exp\{\delta_j\}, \quad (80)$$

where δ_j is a persistent SA2-specific adjustment common across crops.

E.2 Regional Labor Elasticities

Because production is Cobb–Douglas in labor, the first-order condition for labor implies

$$\beta_{jcL} = \frac{w_{jt} L_{jct}}{Y_{jct}}, \quad (81)$$

where w_{jt} is the agricultural wage.

We observe total agricultural labor L_{jt} at the regional level rather than its allocation across

crops. Summing the crop-specific first-order conditions and substituting Equation 80 gives

$$\exp\{\delta_j\} = \frac{w_{jt}L_{jt}}{\sum_c \widehat{\beta}_{cL} Y_{jct}} \quad (82)$$

in the absence of measurement error or transitory deviations from the model.

Define the observed annual regional deviation

$$d_{jt} \equiv \log \left(\frac{w_{jt}L_{jt}}{\sum_c \widehat{\beta}_{cL} Y_{jct}} \right). \quad (83)$$

We allow this object to contain both a persistent regional component and a transitory deviation:

$$d_{jt} = \delta_j + \nu_{jt}, \quad \nu_{jt} \sim \mathcal{N}(0, \sigma_\nu^2). \quad (84)$$

We discipline persistent regional departures from the farm-level estimates with the hierarchical prior

$$\delta_j \sim \mathcal{N}(0, \sigma_\delta^2). \quad (85)$$

Centering the distribution at zero centers the regional elasticity on Rafey's farm-level estimate while allowing the data to reveal persistent departures from it.

Let

$$\bar{d}_j = \frac{1}{T_j} \sum_t d_{jt} \quad (86)$$

denote the average observed deviation for region j . Conditional on the variance components, normality implies

$$\delta_j \mid \bar{d}_j \sim \mathcal{N} \left(\kappa_j \bar{d}_j, \left[\frac{1}{\sigma_\delta^2} + \frac{T_j}{\sigma_\nu^2} \right]^{-1} \right), \quad (87)$$

where

$$\kappa_j = \frac{\sigma_\delta^2}{\sigma_\delta^2 + \sigma_\nu^2 / T_j}. \quad (88)$$

The Empirical Bayes estimate is therefore the posterior mean

$$\widehat{\delta}_j = \widehat{\kappa}_j \bar{d}_j. \quad (89)$$

Regions for which the observed labor share is measured precisely receive little shrinkage, while noisier regional estimates are pulled more strongly toward the farm-level benchmark. We estimate σ_δ^2 and σ_ν^2 jointly by maximum marginal likelihood. For region j , stack the T_j annual signals into $\mathbf{d}_j = (d_{j1}, \dots, d_{jT_j})'$. Integrating out the persistent regional component δ_j gives

$$\mathbf{d}_j \sim N(0, \Sigma_j), \quad \Sigma_j = \sigma_\nu^2 I_{T_j} + \sigma_\delta^2 \mathbf{1}_{T_j} \mathbf{1}_{T_j}'. \quad (90)$$

We estimate the variance components by maximizing the resulting marginal likelihood across re-

gions. Equivalently, region j contributes

$$-\frac{1}{2} \left[T_j \log(2\pi) + (T_j - 1) \log \sigma_\nu^2 + \log(\sigma_\nu^2 + T_j \sigma_\delta^2) + \frac{\sum_t (d_{jt} - \bar{d}_j)^2}{\sigma_\nu^2} + \frac{T_j \bar{d}_j^2}{\sigma_\nu^2 + T_j \sigma_\delta^2} \right] \quad (91)$$

to the log likelihood. The within-region variation in d_{jt} identifies σ_ν^2 , while persistent differences in regional means identify σ_δ^2 .

The resulting regional crop-specific labor elasticities are

$$\hat{\beta}_{jcL} = \hat{\beta}_{cL} \exp\{\hat{\delta}_j\}. \quad (92)$$

E.3 Crop-Region Productivity

Conditional on the regional labor elasticities, crop-region-year productivity is the residual required for Equation 79 to rationalize observed output. Define

$$A_{jct} = \hat{\alpha}_c (W_{jct} + \hat{\theta}_c R_{jct})^{\frac{\hat{\sigma}_c - 1}{\hat{\sigma}_c}} + (1 - \hat{\alpha}_c) K_{jct}^{\frac{\hat{\sigma}_c - 1}{\hat{\sigma}_c}}. \quad (93)$$

Then productivity is recovered by inversion:

$$\hat{\omega}_{jct} = \log Y_{jct} - \frac{\hat{\sigma}_c}{\hat{\sigma}_c - 1} \hat{\beta}_{cW} \log A_{jct} - \hat{\beta}_{jcL} \log L_{jct}. \quad (94)$$

Because agricultural employment is observed at the regional rather than crop level, we allocate regional labor across crops according to the model-implied labor shares,

$$L_{jct} = L_{jt} \frac{\hat{\beta}_{cL} Y_{jct}}{\sum_{c'} \hat{\beta}_{c'L} Y_{jct}}. \quad (95)$$

E.4 Marginal Productive Value of Water

With the production parameters and productivity shifters in hand, we evaluate the marginal product of irrigation water at each observed crop-region-year allocation. Let

$$\rho_c \equiv \frac{\hat{\sigma}_c - 1}{\hat{\sigma}_c} \quad (96)$$

and

$$A_{jct} = \hat{\alpha}_c (W_{jct} + \hat{\theta}_c R_{jct})^{\rho_c} + (1 - \hat{\alpha}_c) K_{jct}^{\rho_c}. \quad (97)$$

Differentiating Equation 79 with respect to irrigation gives

$$\hat{V}_{jct}^P = \frac{\partial Y_{jct}}{\partial W_{jct}} = Y_{jct} \hat{\beta}_{cW} \frac{\hat{\alpha}_c (W_{jct} + \hat{\theta}_c R_{jct})^{\rho_c - 1}}{A_{jct}}. \quad (98)$$

Because output is measured in value terms, this derivative gives the marginal contribution of an additional unit of irrigation to producer revenue and, by the envelope argument in Section 5, to optimized operating profits.

Finally, we aggregate crop-specific marginal values to a single region-year value using observed irrigation shares:

$$\widehat{V}_{jt}^P = \sum_c s_{jct}^W \widehat{V}_{jct}^P, \quad s_{jct}^W = \frac{W_{jct}}{\sum_{c'} W_{jc't}}. \quad (99)$$

This corresponds to valuing a marginal proportional increase in regional irrigation while holding fixed the observed allocation of water across crops.

F Converting Community Water Values to Dollars

Our estimates of community water values are initially in utility units for a 10% change in applied irrigation. We convert these estimates into annual community willingness to pay in AUD per ML.

We first use the coefficient on log weekly housing costs, $\alpha_p < 0$, as the money metric for utility. If weekly housing costs in community j are p_j , one unit of flow utility corresponds to $52p_j/|\alpha_p|$ dollars per household per year. We aggregate across households using population N_j and average household size $\bar{h}$.

We then convert the normalized water measure into applied water. Let s_W denote the change in residual ET, in ML/ha, corresponding to a 10% change in applied irrigation. Our validation of residual ET against administrative data on regional water allocations implies that one additional ML of allocated water generates 0.65 ML of dry-season residual ET. Finally, because residual ET is measured per hectare, we use the mean SA1 area in our estimation sample, $\bar{A}$, to convert the intensity change into a total water quantity. We use a common rather than community-specific area so that geographic differences in administrative-unit size do not mechanically generate differences in per-ML community values.

Combining these conversions gives

$$V_j^C = \omega_j \left(\frac{52p_j}{|\alpha_p|} \right) \left(\frac{N_j}{\bar{h}} \right) \left(\frac{0.65}{s_W \bar{A}} \right), \quad (100)$$

where ω_j is community j 's utility value of a 10% change in applied irrigation. Equation (100) yields community willingness to pay in AUD per ML of applied irrigation water.²³

We apply the same linear transformation to the empirical-Bayes posterior distribution of ω_j . If C_j denotes the conversion factor multiplying ω_j in Equation (100), then

$$\widehat{V}_j^C = C_j \widehat{\omega}_j^{EB}, \quad \text{SD}(V_j^C \mid \text{data}) = |C_j| \text{SD}(\omega_j \mid \text{data}). \quad (101)$$

G Take-It-or-Leave-It Compensation Offers

This appendix derives the take-it-or-leave-it (TIOLI) compensation mechanism used in Section 8. The problem is the procurement analogue of the optimal posted-price problem in Myerson (1981)

²³Housing costs serve only as the numeraire for converting utility into dollars. Changes in housing costs induced by water reallocation are not separately counted as community welfare effects.

and [Bulow & Roberts \(1989\)](#): traders have a known surplus from completing the transaction but face a community whose reservation value is privately known.

Let $S > 0$ denote the surplus available from a potential trade. The selling community privately observes its value of retaining the water, V_s^C , while traders know its population distribution,

$$V_s^C \sim F_C, \quad (102)$$

with density f_C . Traders make a compensation offer p , which the community accepts whenever $V_s^C \leq p$. The probability that the trade is accepted is therefore $F_C(p)$. Conditional on acceptance, traders retain $S - p$, so their expected payoff is

$$\Pi(p; S) = (S - p)F_C(p). \quad (103)$$

The optimal offer solves

$$p^*(S) = \arg \max_{0 \leq p \leq S} (S - p)F_C(p). \quad (104)$$

For an interior solution, the first-order condition is

$$(S - p^*)f_C(p^*) = F_C(p^*), \quad (105)$$

or equivalently,

$$S - p^* = \frac{F_C(p^*)}{f_C(p^*)}. \quad (106)$$

The left-hand side is the surplus traders retain when the offer is accepted. A marginal increase in the offer reduces this surplus on every accepted trade, but increases the probability of acceptance by $f_C(p)$. The optimal offer balances these two effects. As in the monopoly-pricing interpretation of optimal auctions in [Bulow & Roberts \(1989\)](#), traders optimally shade the offer below the amount they could afford to pay because a higher offer increases the probability of trade at the cost of lower surplus conditional on trade. This informational friction can therefore prevent some trades for which $S \geq V_s^C$ and full compensation would have been feasible.

In our implementation, we approximate F_C by a Normal distribution using the estimated population mean and variance of selling-community values and solve Equation (104) numerically for each potential trade. We consider two definitions of available surplus. With private gains alone,

$$S = G_{sb}^P, \quad (107)$$

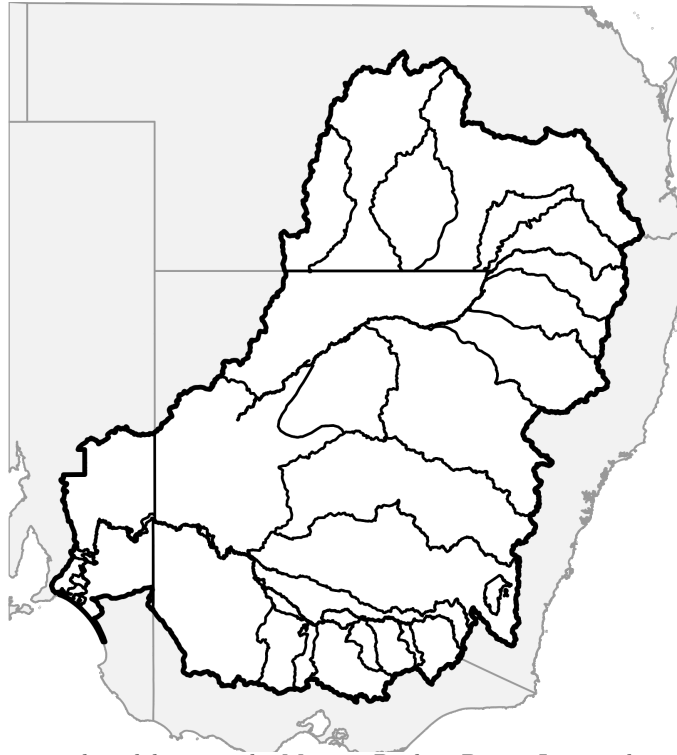
while allowing gains to the buying community to finance compensation gives

$$S = G_{sb}^P + \max\{V_b^C, 0\}. \quad (108)$$

The information available to traders about V_s^C is the same in both cases; buying-community gains increase only the surplus available to finance the offer. A trade is implemented when the realized selling-community value satisfies $V_s^C \leq p^*(S)$, in which case the community receives $p^*(S)$.

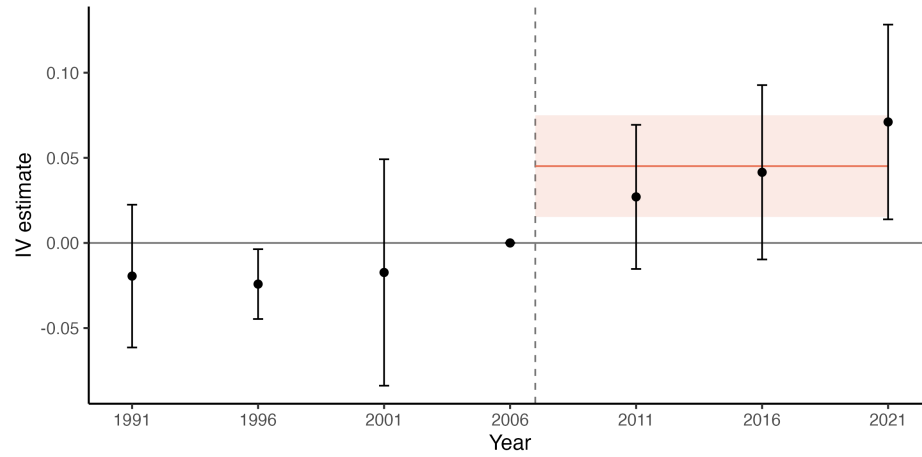
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Figure 8: Murray-Darling Basin Map and Water Districts

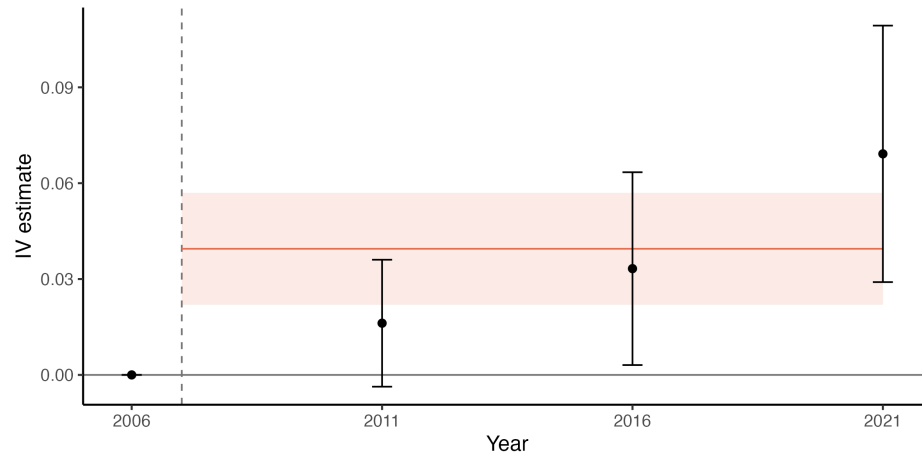


Notes: The heavy outline delineates the Murray–Darling Basin. Interior boundaries show the water districts used to organize the administrative water data. State boundaries are shown in gray for geographic reference.

Figure 9: Effect of Water Gain on Population and Long-Term Residents



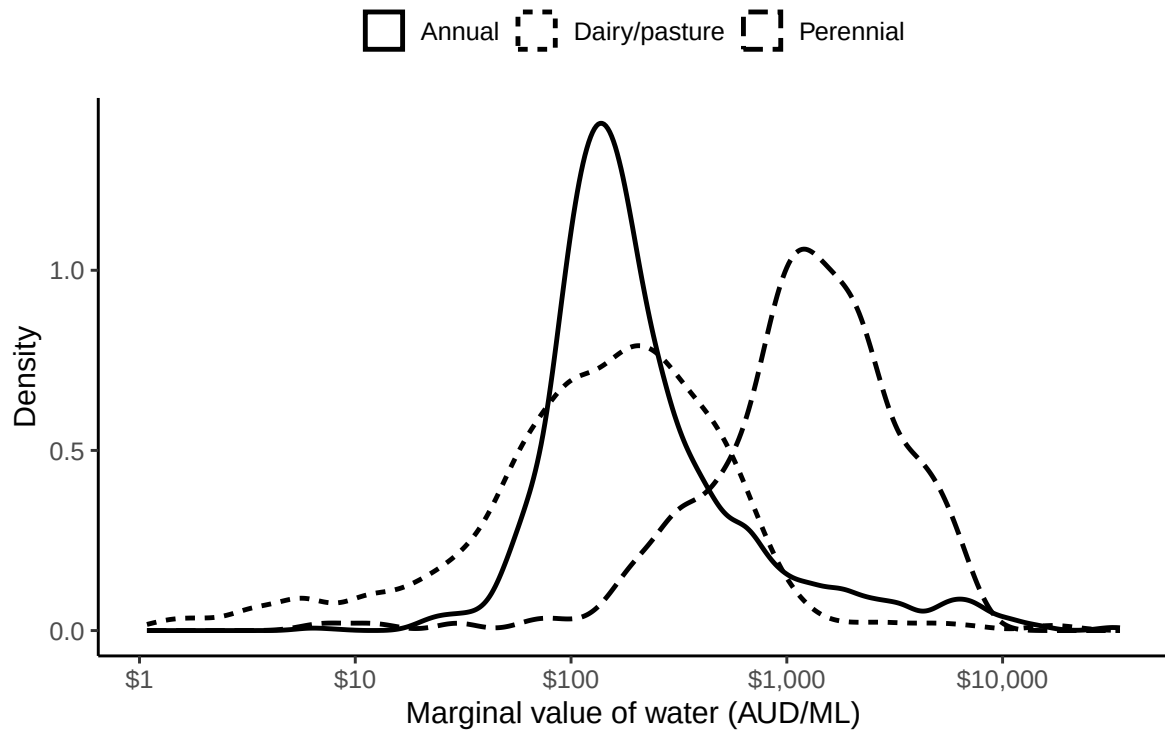
(a) Population



(b) Long-term residents

Notes: Figure depicts the effect of a persistent increase of 0.096 ET/hectare on SA1 population and the stock of long-term residents relative to 2006 for each Census year from 1991–2021 when data is available. We instrument for water change with soil characteristics. One unit of water change is equivalent to a 10% increase in irrigation intensity at the average SA1. Annual estimates are reported in black dots and the pooled post-reform effect is the horizontal red line. Standard errors are clustered at the SA2 level for annual results and at the SA1 level for the pooled effect.

Figure 10: Distribution of the Marginal Productive Value of Water



Notes: The figure plots the distribution of the estimated marginal value of applied irrigation water across crop-SA2-year observations. The horizontal axis is logarithmic. The estimation sample shown here excludes crop-SA2-year cells with fewer than 100 hectares of observed production. Marginal values are measured in AUD per megaliter.

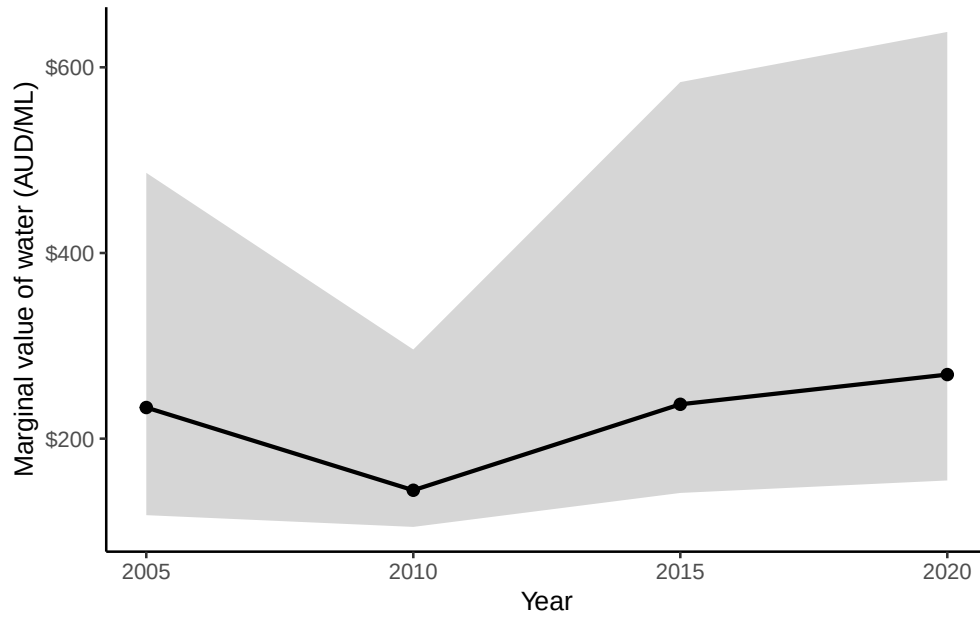
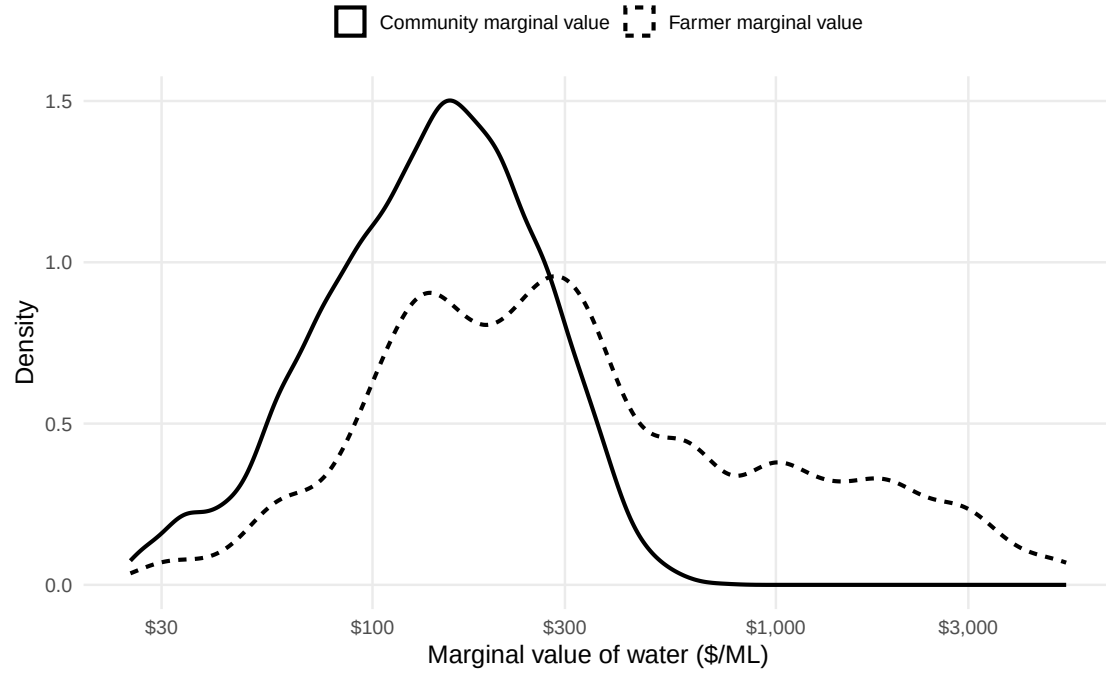


Figure 11: Marginal Productive Value of Water over Time

Notes: The solid line reports the median water-weighted marginal value of irrigation water across SA2s in each year. The shaded region denotes the interquartile range. Crop-specific marginal values are aggregated within an SA2 using observed shares of applied irrigation water. Values are measured in AUD/ML.

Figure 12: Comparing private and community values for water



Notes: Each panel plots the distribution of the water-weighted marginal productive value of irrigation water across SA2s against the marginal community value of water across SA1s. The horizontal axis is logarithmic. Values are measured in AUD/ML.

I Appendix Tables

Table 6: SA1-year Census Variable Summary

Variable	N	Mean	SD	P10	Median	P90
Long-tenure share	23,424	0.5	0.1	0.4	0.5	0.7
Household income	23,152	1,181.9	705.1	454.8	1,029.0	2,154.0
Population	23,782	349.8	147.9	192.0	344.0	523.0
Tenure stock	21,752	0.8	2.9	0.5	0.7	1.0
Weekly mortgage payment	23,094	1,323.3	586.1	699.3	1,300.0	2,059.0
Weekly rent	23,114	222.0	128.7	75.0	203.0	395.0

Notes: *Notes:* The table summarizes the SA1-year observations used to construct local outcomes and residential characteristics. N is the number of nonmissing SA1-year observations for each variable; SD is the standard deviation, and P10 and P90 are the 10th and 90th percentiles. Income, rent, and mortgage payments are reported in Australian dollars in their underlying Census reporting periods. Long-tenure share and tenure stock are proportions.

Table 7: First Stage: Agricultural Suitability and Water Reallocation

	2011	2016	2021	Pooled Post
Root-zone water storage	-0.0066*** (0.0011)	-0.0055*** (0.0009)	-0.0052*** (0.0010)	-0.0057*** (0.0004)
Water-region fixed effects	Yes	Yes	Yes	No
Water-region $\times$ year fixed effects	No	No	No	Yes
Observations	6,494	6,494	6,494	19,482
Excluded-instrument F -statistic	34.1	35.6	28.0	183.9

Notes: The dependent variable in each column is the change in local irrigation activity accumulated through the indicated year relative to the pre-reform period. Root-zone water storage is constructed from predetermined soil characteristics. Columns (1)–(3) include water-region fixed effects and report standard errors clustered by SA2 in parentheses. Column (4) pools the three post-reform years, includes water-region-by-year fixed effects, and clusters standard errors by SA1. The excluded-instrument F -statistic is the clustered Wald F -statistic for root-zone water storage.

Table 8: Effects of Water Reallocation on Local Outcomes

Outcome	Pooled estimate	95% confidence interval
<i>Economic Activity</i>		
Household Income	0.0306 (0.0056)	[0.0196, 0.0415]
Rent	0.0516 (0.0089)	[0.0341, 0.0690]
Mortgage	0.0513 (0.0088)	[0.0341, 0.0685]
Personal Income	0.0030 (0.0028)	[-0.0024, 0.0084]
Ag. Employment	0.0750 (0.0189)	[0.0380, 0.1120]
Total Businesses	0.0261 (0.0054)	[0.0156, 0.0366]
<i>Population and residence</i>		
Population	0.0451 (0.0152)	[0.0153, 0.0750]
Tenure Stock	0.0368 (0.0083)	[0.0204, 0.0532]
<i>Political and social outcomes</i>		
Labor Vote Share	-0.0215 (0.0103)	[-0.0417, -0.0013]
Suicide Rate	-0.1736 (0.0971)	[-0.3640, 0.0167]

Notes: The table reports pooled post-reform IV estimates of the effect of a persistent 0.096 ML/ha increase in dry-season residual evapotranspiration (ET). Water changes are instrumented using predetermined soil characteristics. This change in residual ET corresponds to approximately a 10% increase in irrigation intensity at the average SA1. Business outcomes, Labour vote share, and the suicide rate are observed at the SA2 level; all other outcomes are observed at the SA1 level. Standard errors, in parentheses, are clustered at the geographic level at which each outcome is observed.

Table 9: Cross-Period Covariance of Community Utility Changes

Residual cross-year covariance	
<i>Variables</i>	
Exposure product	0.145*** (0.052)
<i>Fixed-effects</i>	
Cross-year	Yes
<i>Fit statistics</i>	
Observations	13,487

Notes: The dependent variable is the product of residual changes in community flow utility in two distinct post-reform periods, after removing the mean effect of irrigation. The exposure product is the product of the corresponding reform-induced changes in local irrigation. The specification pools the 2011–2016, 2011–2021, and 2016–2021 period pairs and includes period-pair fixed effects. Under Equation 30, its coefficient estimates the variance of persistent community-specific effects of irrigation, σ_{η}^2 .

Table 10: Agricultural Production Given Parameters

Crop	α_c	σ_c	θ_c	β_{cW}	β_{cL}
Perennial	0.599	1.575	1.081	0.631	0.352
Annual	0.513	1.451	1.048	0.526	0.201
Dairy	0.385	5.000	0.148	0.782	0.147

Notes: The table reports crop-specific production parameters taken from [Rafey \(2023\)](#) and held fixed when applying the production function to our SA2 data. α_c is the share parameter for effective water in the water–land aggregate; σ_c is its elasticity of substitution; θ_c converts rainfall into effective water; and β_{cW} and β_{cL} are the output elasticities of the water–land aggregate and labor. For annual crops, we use the parameters estimated for irrigated production because our objective is to value an additional unit of applied irrigation at observed irrigated allocations.

Table 11: Distribution of Recovered Agricultural Productivity

crop	N	Mean	SD	p10	Median	p90
annual	8217	10.711	4.879	8.191	11.300	13.476
dairy	8298	7.009	3.747	4.640	7.468	9.534
perennial	5364	8.877	6.626	6.690	9.435	11.669

Notes: The table summarizes the crop–SA2–year revenue productivity shifter $\widehat{\omega}_{jct}$ recovered by inverting the production function at observed output and inputs. The shifter absorbs differences in output prices and unobserved intermediate inputs as well as physical productivity; its levels should therefore not be interpreted as comparable measures of physical productivity across crops. N counts crop–SA2–year observations.

Table 12: Agricultural Production Estimated Parameters

	Mean	SD	p25	Median	p75
$\exp\{\widehat{\delta}_j\}$	0.727	1.119	0.356	0.652	1.273

Notes: The table summarizes $\exp\{\widehat{\delta}_j\}$ across SA2s. This factor multiplies the crop-specific farm-level labor elasticity taken from [Rafey \(2023\)](#) to obtain the effective regional labor elasticity, $\widehat{\beta}_{jcl} = \widehat{\beta}_{cL} \exp\{\widehat{\delta}_j\}$. A value of one leaves the farm-level elasticity unchanged; values below or above one imply a lower or higher regional elasticity, respectively. SD is the standard deviation, and P25 and P75 are the 25th and 75th percentiles.

Table 13: Distribution of the Marginal Productive Value of Water

Crop	N	Mean	p25	Median	p75
Annual	2016	597	111	167	323
Dairy/pasture	2349	344	64	153	319
Perennial	621	1735	705	1258	2285
All crops, water-weighted	705	565	122	215	509

Notes: Entries report summary statistics for the estimated marginal value of applied irrigation water, in AUD/ML. Crop-specific statistics are calculated across crop-SA2-year observations with at least 100 hectares of production. The final row aggregates crop-specific marginal values within each SA2-year using each crop's share of applied water as weights.

Table 14: Administrative Water Components and Residual Evapotranspiration

Table 15: Regional Residual ET versus Allocation Decomposition

Residual ET Model:	Annual (1)	Dry (2)	Wet (3)
<i>Variables</i>			
Direct allocation	0.7268* (0.4073)	0.6518* (0.3380)	-0.0387 (0.1354)
Environmental buyback	-0.5730 (3.743)	-0.6834 (2.262)	0.0996 (1.016)
Net agricultural transfer	0.7336 (0.9230)	0.5954 (0.6637)	0.0970 (0.3374)
<i>Fixed-effects</i>			
region_id	Yes	Yes	Yes
year	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	440	440	440
R ²	0.31990	0.78178	0.93051

Notes: The dependent variables are annual, dry-season (October–March), and wet-season residual evapotranspiration, respectively. Administrative water availability is decomposed into direct allocations, environmental buybacks, and net agricultural transfers. All specifications include water-region and year fixed effects. Standard errors are clustered at the water-region level.

Table 16: Validation of Residual ET with Applied Irrigation Water

Dependent Variable:	applied_water		
	Applied Water		
Model:	(1)	(2)	(3)
<i>Variables</i>			
Residual ET	1.19*	1.68**	-1.10
	(0.693)	(0.843)	(1.83)
<i>Fixed-effects</i>			
sa2	Yes	Yes	Yes
water_id-year	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	1,170	1,170	1,170
R ²	0.720	0.721	0.719

Notes: The dependent variable is administrative applied irrigation water at the SA2-year level. Columns report specifications using annual, dry-season (October–March), and wet-season residual evapotranspiration, respectively. All specifications include SA2 fixed effects and water-district-by-year fixed effects. Standard errors are clustered at the SA2 level.